INSTITUTE FOR CONTINUNING AND INNOVATIVE EDUCATION
KANO UNIVERSITY OF SCIENCE AND TECHNOLOGY, WUDIL
P.M.B. 3244 KANO www.iciekustwudil.edu.ng

2020/2021 COURSE MATERIAL FOR MTH 1304
(CALCULUS AND ITS APPLICATIONS)

PREPARED BY

AUGUST, 2021

General Information for Students

1. **Course Description / Content**

Function of variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation. Extreme curve sketching; integration as an inverse of differential. Methods of integration, definite integrals. Application to areas and volume.

This note is divided into three sections and throughout this course we are going to consider the set of real numbers, denoted by $\mathbb{R}$

2. Course Status

MTH 1304 is a Two (2) credit Core (Compulsory) Course to all level 100 students pursuing B. Agriculture (Hons) part time degree programme of Kano University of Science and Technology, Wudil.

3. Reference Materials

(a) ICIE MTH 1304 Course Material

(b) Any other relevant / recommended text book

4. Assessment

(a) Continuous Assessment Attendance – 10 %

(b) Continuous Assessment Assignment (maximum of two) – 15 %

(c) Continuous Assessment Test – 15%

(d) Semester Examination – 60%

(e) Continuous Assessment (C.A) = Attendance + Assignment + Test

= 10% + 15% + 15% = 40%

(vi) Total Percentage Score = Continuous Assessment (C.A.) + Semester Examination

= 40% + 60% = 100%

5. Eligibility for Examination – Students are expected to have 75% to be eligible for semester examination

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SECTION A

Functions, Limit and Continuity

1.1 Functions

A function f is a rule that assigns to each element $x \in X \subseteq \mathbb{R}$ a single element $f(x) \in Y \subseteq \mathbb{R}$

Remark 1.1

(i) The set X is called the **domain** of the function, written as $dom\ f$. It consists of all x such that $f(x)$ is defined.

(ii) The number $f(x)$ is the **value of f at x** and is read "f of x." For example, if $f(x) = 2x - 1$, the value of $f(x)$ at $x = 2$ is $f(2) = 3$.

(iii) The **range** of f is the set of all possible values of $f(x)$ as varies throughout the domain.

(iv) We are going to represent a function algebraically (i.e., by an explicit formula). For example, $f(x) = x + 1, g(x) = x^2 + 2x - 1$ are functions.

For those functions which are represented algebraically, we can determine their domains and ranges as in the following examples.

Example 1.1

Find the domain and range of each function (a) $f(x) = 2x + 1$ (b) $g(x) = \sqrt{x+2}$

(c) $f(x) = \frac{1}{x^2 - x}$ (d) $h(x) = \begin{cases} 2, & if\ x \geq 0 \\ 5, & if\ x < 0 \end{cases}$

Solution

(a) Since f is defined for any real number, then the domain of f is the set of real number. That is, dom $f = \mathbb{R}$.

The range is a collection of all images which are real numbers and so, Range $f = \mathbb{R}$.

(b) The domain of g consists of all values of x such that $x + 2 \geq 0$ which is equivalent to $x \geq -2$. Hence the domain is the interval $[-2, \infty)$. The range of g is $(0, \infty)$.

(c) $f(x) = \frac{1}{x^2 - x} = \frac{1}{x(1-x)}$ which is not defined when $x = 0$ or $x = 1$ since division by 0 is not allowed. Thus, the domain of g is the set $\{x: x \neq 0, x \neq 1\}$ or equivalently $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$.

Range $f = (-\infty, 0) \cup (0, \infty)$

(d) Since h is defined for both non negative and negative real numbers, dom $h = \mathbb{R}$.

The range of h is $\{2, 5\}$.

Exercise:

1. Find the domain of the function (i) $f(x) = \frac{x+4}{x^2 - 9}$ (ii) $g(x) = \sqrt{2 - \sqrt{p}}$

(iii) $h(x) = |x| - x$ (iv) $F(x) = \begin{cases} x+2 & if\ x < 0 \\ 1-x & if\ x \geq 0 \end{cases}$

2. Find the domain and range of the function

(a) $f(x) = \sqrt{4-x^2}$ (b) $g(x) = \begin{cases} -1 & if\ x \leq -1 \\ 3x+2 & if\ |x| < 1 \\ 7-2x & if\ x \geq 1 \end{cases}$

1.2 Some Special functions

1.2.1 Constant function

A function which assigns a fixed value for every value of x is called a constant function.

Example, $f(x) = 3x^0$, $g(x) = 5$.

1.2.2 Absolute value function

The **absolute value** of a number x, denoted by $|x|$, is the distance from x to 0 on the real number line. Distances are always positive or 0, so $|x| \geq 0$ for every real number x. In general

$$|x| = \begin{cases} x, & if\ x \geq 0 \\ -x, & if\ x < 0 \end{cases}$$

1.2.3 Even and odd functions

An **even function *f*** is a function for which $f(-x) = f(x)$ for every number x in its domain. For instance, the function $f(x) = x^2$ is even because

$$f(-x) = (-x)^2 = (-1)^2x^2 = x^2 = f(x)$$

An odd function satisfies $f(-x) = -f(x)$ for every number x in its domain. For example, $f(x) = x^3$ is an odd function since

$$f(-x) = (-x)^3 = (-1)^3x^3 = -x^3 = -f(x)$$

Example 1.2.3 Determine whether each of the following functions is even, odd, or neither even nor odd. (a) $f(x) = x^5 + x$ (b) $g(x) = 1 - x^4$ (c) $h(x) = 2x - x^2$

Solution

(a) $f(-x) = (-x)^5 + (-x) = -(x^5 + x) = -f(x)$. This is an odd function

(b) $g(-x) = 1 - (-x)^4 = 1 - x^4 = f(x)$. This is an even function

(a) $h(-x) = 2(-x) - (-x)^2 = -2x - x^2 = -(2x + x^2)$. This is neither odd nor even

1.2.4 Increasing and decreasing functions

A function f is said to be increasing on an interval *I* if

$$f(x_1) < f(x_2) \text{ whenever } x_1 < x_2 \text{ in } I$$

It is decreasing on *I* if

$$f(x_1) > f(x_2) \text{ whenever } x_1 > x_2 \text{ in } I$$

For example, the function $f(x) = x^2$ is a decreasing function on the interval $(-\infty, 0]$ and increasing on the interval $[0, \infty)$.

1.3 Combinations of functions

Let f and g be two functions, then we can combine them to form new functions $f + g, f - g, fg$, and $\frac{f}{g}$ in a similar way we add, subtract, multiply and divide real numbers. The sum and difference functions are defined by

$$(f + g)(x) = f(x) + g(x), \quad (f - g)(x) = f(x) - g(x).$$

Similarly, the product and quotient functions are defined by

$$(fg)(x) = f(x)g(x), \text{ and } \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Remark 1.3

(i) If the domain of f is A and the domain of g is B, then the domain of $f + g, f - g$, and $fg$is the intersection $A \cap B$ because both f and g have to be defined. For example, the domainof $f = \sqrt{x}$ is $A = [0, \infty)$ and the domain of $g(x) = \sqrt{2 - x}$ is $B = (-\infty, 2]$, so the domain of $f + g = \sqrt{x} + \sqrt{2 - x}$ is $A \cap B = [0, 2]$..

(ii) Since division by 0 is not allowed, the domain of $\frac{f}{g}$ is $\{x \in A \cap B | g(x) \neq 0\}$. For instance, if $f(x) = x^2$ and $g(x) = x - 1$, then the domain of the function $\left(\frac{f}{g}\right)(x) = \frac{x^2}{x-1}$ is $\{x | x \neq 1\}$ or $(-\infty, 1) \cup (1, \infty)$.

1.3.1 Composition of functions

This is another way of combining two functions to get a new function. Let $f: X \to Y$ and $g: Y \to Z$ be two functions, the composite function $f \circ g$ is a rule that assigns to each $x \in X$ a unique $z = f\big(g(x)\big) \in Z$. It is denoted by

$$(f \circ g)(x) = f\big(g(x)\big).$$

The composite function $(f \circ g)(x)$ which is also called the composition of f and g is defined whenever both $g(x)$ and $f(g(x))$ are defined.

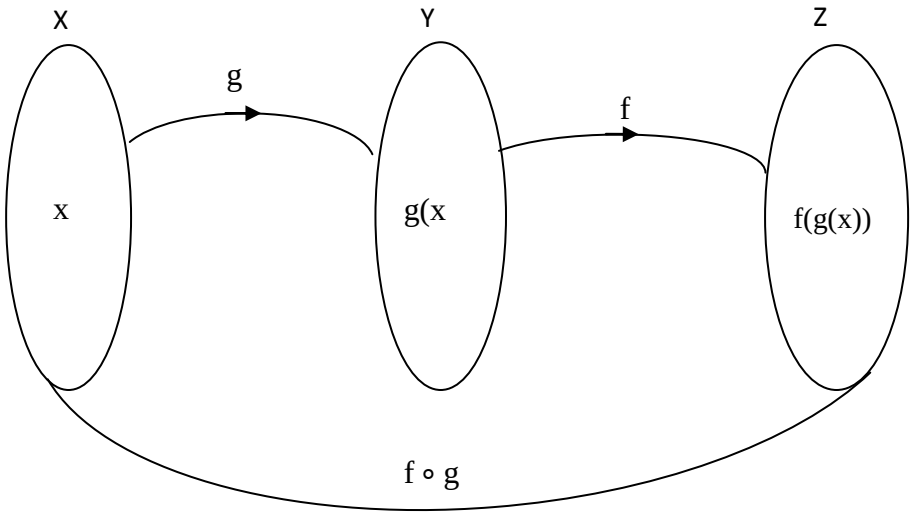


Figure 1. Composition of *f* and *g*

Example 1.3.1 Let $f(x)=x^2$ and $g(x)=x-3$. Find the composite functions $f \circ g$ and $g \circ f$

Solution

$$(f \circ g)(x) = f\big(g(x)\big) = f(x-3) = (x-3)^2$$

$$(g \circ f)(x) = g\big(f(x)\big) = g(x^2) = x^2 - 3$$

Example 1.3.2 If $f = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find each function and its domain

(a) $f \circ g$ (b) $g \circ f$ (c) $f \circ f$ (d) $g \circ g$

Solution

(a) $(f \circ g)(x) = f\big(g(x)\big) = f\big(\sqrt{2-x}\big) = \sqrt[4]{2-x}$

The domain of $f \circ g$ is $\{x|2-x \geq 0\} = \{x|x \leq 2\} = (-\infty, 2]$

(b) $(g \circ f)(x) = g\big(f(x)\big) = g\big(\sqrt{x}\big) = \sqrt{2-\sqrt{x}}$

For both $\sqrt{x}$ and $2-\sqrt{x}$ to be defined, we require $x \geq 0$ and $2-\sqrt{x} \geq 0$ respectively. That is

$\sqrt{x} \leq 2$ or $x \leq 4$. Thus, $0 \leq x \leq 4$ and so, the domain of $g \circ f$ is $[0, 4]$.

(c) $(f \circ f)(x) = f\big(f(x)\big) = f\big(\sqrt{x}\big) = \sqrt{\sqrt{x}} = \sqrt[4]{x}$. Thus, the domain of $f \circ f$ is $[0, \infty)$

(d) $(g \circ g)(x) = g\big(g(x)\big) = g\big(\sqrt{2-x}\big) = \sqrt{2-\sqrt{2-x}}$

The expression $\sqrt{2-\sqrt{2-x}}$ is defined when $2-x \geq 0$ or $x \leq 2$ and $2-\sqrt{2-x} \geq 0$ or $x \geq -2$. Thus, the domain of $g \circ g$ is $[-2, 2]$.

Remark 1.3.1

It is possible to make a composition of more than two functions. For example, the composite function $f \circ g \circ h$ is found by first applying h, then g and then f as follows:

$$(f \circ g \circ h)(x) = f\Big(g\big(h(x)\big)\Big).$$

Exercise:

1. Determine whether f is even, odd, or neither

   (i) $f(x) = \frac{x}{x^2+1}$ (ii) $f(x) = 1 + 3x^2 - x^4$ (iii) $f(x) = x|x|$

2. Find (a) $f + g$ (b) $f - g$ (c) fg and (d) $\frac{f}{g}$ and state their domains

   $f(x) = x^3 + 2x^2,\ g(x) = 3x^2 - 1$ and $f(x) = \sqrt{3-x},\ g(x) = \sqrt{x^2-1}$

3. Find the functions $f \circ g,\ g \circ f,\ f \circ f$, and $g \circ g$ and their domains

   (a) $f = x^2 - 1, g(x) = 2x + 1$ (b) $f(x) = 1 - 3x,\ g(x) = cosx$

   (c) $f(x) = \sqrt{x},\ g(x) = \sqrt[3]{1-x}$ (d) $f(x) = x + \frac{1}{x}, g(x) = \frac{x+1}{x+2}$

1.4 Graph of a function

The graph of a function consists of ordered pairs $(x,\ y)$ such that $y = f(x)$. The process of sketching the graph of a function is called graphing the function. The following three steps are taken when graphing a function.

(i) Make a table of pairs from the functions.

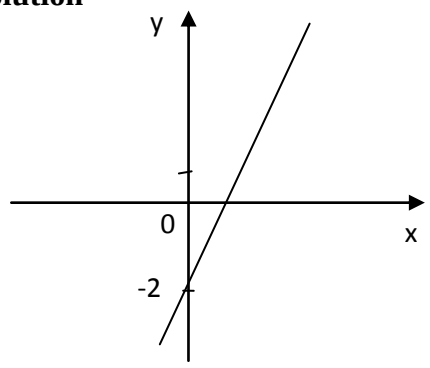
(ii) Plot enough of the corresponding points to learn the shape of the graph. Add more points to the table if necessary.

(iii) Complete the sketch by connecting the points.

Example 1.4

(1) Graph the function $f(x) = 3x - 2, -2 \leq x \leq 2$

Solution



X	0.0	1.0	1.1	1.2	1.3	1.4	1.5
f(x)	-2.0	1.0	1.3	1.6	1.9	2.2	2.5

1.5 Limit of a function

To define a limit of a function intuitively, let us investigate the behavior of the function $f(x)$ defined by $f(x) = x^2 + 1$ for values of x near 2. The table below shows the values of $f(x)$ for values of x close to 2, but not equal to 2.

Table 1

x	$f(x)$		x	$f(x)$
1	2		3	10.00
1.5	3.25		2.5	6.29
1.7	3.89		2.3	5.41
1.9	4.61		2.1	5.2025
1.95	4.8025		2.05	5.0401
1.99	4.9601		2.01	5.004001
1.999	4.99001		2.001	5.0040001
1.9999	4.99960001		2.0001	5.00040001

From the above table we can see that when *x* gets close to 2 (from either side of 2), $f(x)$ is close to 5. Thus, we can make the values of $f(x)$ as close as we like to 5 by taking x sufficiently close to 2. This can be expressed by saying that " the limit of the function $f(x)$ as x approaches 2 is 5." By notation, we have $\lim_{x\to 2}(x^2 + 1) = 5$.

Definition 1.5.1:- Let $f(x)$ be a function defined when x is near the number k. That is, f is defined on some open interval that contains k, except possibly at k itself. Then

$$\lim_{x\to k} f(x) = L,$$

for some real number L and say " the limit of $f(x)$, as x approaches k, equals L.

Remark 1.5.1

An alternative notation for $\lim_{x\to k} f(x) = L$, is $f(x) \to L$ as $x \to k$ which usually read "$f(x)$ approaches L as x approaches k."

Example 1.5.1 Find the limit of the function $f(x) = x^2 + 3x + 4$ as $x \to 2$.

Solution

$$\lim_{x\to 2} x^2 + 3x + 4 = (2^2 + 3 \times 2 + 4) = 14$$

Example 1.5.2 Compute the limit as $x \to 1$ of $f(x) = \frac{x^2-1}{x-1}$

Solution

Since $f(x)$ is not defined at $x = 1$ and $f(x) = \frac{x^2-1}{x-1} = x + 1,$ then $\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1} x + 1 = 2.$

1.5.1 Calculating limits

The following properties of limits, called limit laws are used to calculate limits

Suppose that c is a real number and the limits

$$\lim_{x\to k} f(x) \text{ and } \lim_{x\to k} g(x)$$

exist. Then

1. The limit of a sum is the sum of limits

$$\lim_{x\to k}[f(x) + g(x)] = \lim_{x\to k} f(x) + \lim_{x\to k} g(x)$$

2. The limit of a difference is the difference of limits

$$\lim_{x\to k}[f(x) - g(x)] = \lim_{x\to k} f(x) - \lim_{x\to k} g(x)$$

3. The limit of a constant times a function is the constant times the limit of the function

$$\lim_{x\to k} cf(x) = c \lim_{x\to k} f(x)$$

4. The limit of a product is the product of the limits

$$\lim_{x\to k}[f(x) \times g(x)] = \lim_{x\to k} f(x) \times \lim_{x\to k} g(x)$$

5. The limit of a quotient is the quotient of the limits provided that the limit of the denominator is not zero

$$\lim_{x\to k} \frac{f(x)}{g(x)} = \frac{\lim_{x\to k} f(x)}{\lim_{x\to k} g(x)} \text{ if } g(x) \neq 0$$

6. Using the product law repeatedly, we have

$$\lim_{x\to k}[f(x)]^n = \left[\lim_{x\to k} f(x)\right]^n,$$

Where n is a positive integer

7. (a) $\lim_{x\to k} c = c$ and (b) $\lim_{x\to k} x = k$

8. (a) Putting $f(x) = x$ in law 6, we have

$$\lim_{x\to k} x^n = k^n.$$

(b) A similar limit holds for roots

$$\lim_{x\to k} \sqrt[n]{x} = \sqrt[n]{k}.$$

(c) In general,

$$\lim_{x\to k} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to k} f(x)},$$

if n is even, it is assumed that $\lim_{x\to k} f(x) > 0.$

Example 1.5.3 Evaluate the following limits and justify each step

a) $\lim_{x\to 5} 2x^2 + 3x - 4$ b) $\lim_{x\to -2} \frac{x^3-2x^2+1}{5-3x}$

Solution

a) $\lim_{x\to 5} 2x^2 + 3x - 4 = \lim_{x\to 5} 2x^2 + \lim_{x\to 5} 3x - \lim_{x\to 5} 4$ (by laws 1 and 2)

$= 2\lim\limits_{x\to 5} x^2 + 3\lim\limits_{x\to 5} x - \lim\limits_{x\to 5} 4$ (by law 3)

$= 2(5)^2 + 3(5) - 4$ (by 8(a), 7(b), 7(a))

$= 61$

b) $\lim_{x\to -2} \frac{x^3-2x^2+1}{5-3x} = \frac{\lim_{x\to -2}(x^3-2x^2+1)}{\lim\limits_{x\to -2}(5-3x)}$ (by law 5)

$= \frac{\lim\limits_{x\to -2} x^3 - 2\lim\limits_{x\to -2} x^2 + \lim\limits_{x\to -2} 1}{\lim\limits_{x\to -2} 5 - 3\lim\limits_{x\to -2} x}$ (by laws 2, 3 and 1)

$= \frac{(-2)^3 - 2(-2)^2 + 1}{5 - 3(-2)}$ (by 8(a), 7(b), 7(a))

$= \frac{-8 - 8 + 1}{5 + 6} = \frac{1}{11}$

9. Direct substitution property: If f is a polynomial or rational function and k is in the domain of f, then

$$\lim_{x\to k} f(x) = f(k).$$

Example 1.5.4 (a) $\lim_{x\to -1} x^5 - 2x^3 + 3x - 1$ (b) $\lim_{x\to 2} \frac{x^3-x^2}{x^2-2x+4}$

Solution

(a) Since $x^5 - 2x^3 + 3x - 1$ is a polynomial then substitution property holds and so,

$$\lim_{x\to -1} x^5 - 2x^3 + 3x - 1 = (-1)^5 - 2(-1)^3 + 3(-1) - 1 = -1 + 2 - 3 - 1 = -3$$

(b) Since $\frac{x^3-x^2}{x^2-2x+4}$ is a rational function, we need to check if the denominator $g(x) = x^2 - 2x + 4$ is not zero when $x = 2$. In fact, $g(2) = 2^2 - 2(2) + 4 = 4 \neq 0$. Therefore,

$$\lim_{x\to 2} \frac{x^3 - x^2}{x^2 - 2x + 4} = \frac{\lim\limits_{x\to 2} x^3 - x^2}{\lim\limits_{x\to 2} x^2 - 2x + 4} = \frac{2^3 - 2^2}{2^2 - 2(2) + 4} = \frac{8 - 4}{4 - 4 + 4} = \frac{4}{4} = 1$$

1.5.2 One-sided limits

Some limits are best calculated by first finding the left- and right-hand limits as a two-sided limit exists if and only if both of the one-sided limits exist and are equal.

Theorem 1.5.2 $\lim_{x\to k} f(x) = L$ if and only if $\lim_{x\to k^-} f(x) = L = \lim\limits_{x\to k^+} f(x)$.

When computing one-sided limits, we use the fact that the Limit Laws also hold for these limits.

Example 1.5.5 Show that $\lim\limits_{x\to 0} |x| = 0$.

Solution

Recall that the absolute value of x is defined by

$$|x| = \begin{cases} x & if\ x \geq 0 \\ -x & if\ x < 0 \end{cases}$$

Since $|x| = x$ for $x > 0$, then $\lim_{x\to 0^+} |x| = \lim_{x\to 0^+} x = 0$

For $x < 0$, $|x| = -x$ so that $\lim_{x\to 0^-} |x| = \lim_{x\to 0^-} (-x) = 0$. Thus $\lim_{x\to 0} |x| = 0$

Example 1.5.6 Prove that $\lim_{x\to 0} \frac{|x|}{x}$ does not exist

Solution

$$\lim_{x\to 0^+} \frac{|x|}{x} = \lim_{x\to 0^+} \frac{x}{x} = 1$$

and

$$\lim_{x\to 0^-} \frac{|x|}{x} = \lim_{x\to 0^+} \frac{-x}{x} = \lim_{x\to 0^+} (-1) = -1$$

Since the right- and left-hand limits are not equal, then it follows from Theorem 1.5.2 that $\lim_{x\to 0} \frac{|x|}{x}$ does not exist

1.5.3 Calculating trigonometric limits

To calculate the limit of trigonometric functions, we need the following result.

Theorem1.5.3 If θ is measured in radians then $\lim_{\theta\to 0} \frac{sin\theta}{\theta} = 1$.

Proof

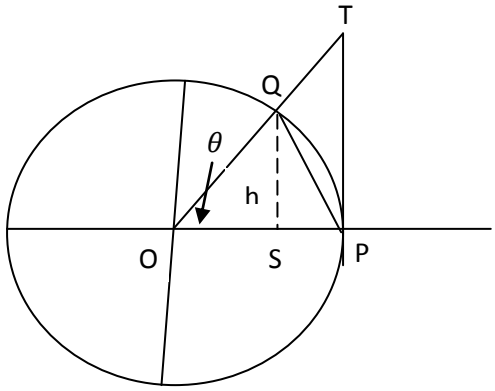


Figure 2

In ΔOTP, $\tan\theta = \frac{TP}{OP}$ implying that TP= $\tan\theta$ and in ΔOQS, $\sin\theta = \frac{h}{OQ} = 1$so that $h = sin\theta$.

But ΔOQP< sector OQP <ΔOTP .(i)

Area of ΔOQP $= \frac{1}{2}|OP|\sin\theta = \frac{1}{2}|x|\sin\theta$ (ii)

Area of sector OQP$= \frac{1}{2}r^2\theta$ (iii)

Area of $\Delta OTP = \frac{1}{2}|OP||TP| = \frac{1}{2}|x|\tan\theta = \frac{1}{2}\tan\theta$ (iv)

Substituting (ii), (iii), (iv) into (i), we have

$\frac{1}{2}|x|\sin\theta < \frac{1}{2}r^2\theta < \frac{1}{2}\tan\theta$

Dividing through by $\frac{1}{2}\sin\theta$, we have

$$\frac{1}{2}|x|\sin\theta < \frac{1}{2}r^2\theta < \frac{1}{2}\tan\theta \Rightarrow |x| < \frac{r^2\theta}{\sin\theta} < \frac{\tan\theta}{\sin\theta}$$

It follows that

$$1 < \frac{\theta}{\sin\theta} < \frac{1}{\cos\theta}$$

Taking the reciprocal, we have

$$1 < \frac{\sin\theta}{\theta} < cos\theta$$

Using $\lim_{\theta\to 0} 1 = 1$ and $\lim_{\theta\to 0} cos\theta = 1$, we obtain

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$$

by the Sandwich theorem which says that if $f(t) \le g(t) \le h(t)$ when *x* is near *k* (except possibly at *k*) and $\lim_{x\to k} f(x) = \lim_{x\to k} h(x) = l$, then $\lim_{x\to k} f(x) = l$.

Example 1.5.7 (i) Calculate $\lim_{x\to 0}\frac{sin5x}{2x}$ (ii) Show that $\lim_{x\to 0}\frac{cos\theta-1}{\theta} = 0$

Solution

(i) To apply Theorem 1.5.3, we need to rewrite the function in such a way that 5x appears in the denominator. That is, $\frac{sin5x}{2x} = \frac{5sin5x}{2(5x)}$ and so

$$\lim_{x\to 0}\frac{sin5x}{2x} = \lim_{x\to 0}\frac{5sin5x}{2(5x)} = \frac{5}{2}\lim_{x\to 0}\frac{sin5x}{5x}.$$

Then by Theorem 1.5.3, we get

$$\lim_{x\to 0}\frac{sin5x}{2x} = \frac{5}{2}.$$

(ii) $\lim_{\theta\to 0}\frac{\cos\theta-1}{\theta} = \lim_{\theta\to 0}\frac{\cos\theta-1}{\theta}\cdot\frac{\cos\theta+1}{\cos\theta+1}$

$$= \lim_{\theta\to 0}\frac{\cos^2\theta-1}{\theta(\cos\theta+1)}$$

$$= \lim_{\theta\to 0}\frac{-\sin^2\theta}{\theta(\cos\theta+1)}, \text{ since } \sin^2\theta+\cos^2\theta = 1$$

$$= -\lim_{\theta\to 0}\left(\frac{\sin\theta}{\theta}\cdot\frac{\sin\theta}{\cos\theta+1}\right)$$

$$= -\lim_{\theta\to 0}\frac{\sin\theta}{\theta}\cdot\lim_{\theta\to 0}\frac{\sin\theta}{\cos\theta+1} = -1\cdot\frac{0}{2} = 0$$

1.5.4 Limits involving infinity

1.5.4.1 Infinite limits

We can see by direct substitution that the limit as $x \to 0$ of $f(x) = \frac{1}{x^2}$ does not exist. But we can observe from the table of values below that the values of $f(x)$ can be made arbitrarily large by taking x close enough to 0.

Table 2

x	*f(x)*	*x*	*f(x)*
1	1	- 1	1
0.5	4	-0.5	4
0.1	100	-0.1	100
0.01	10 000	-0.01	10 000
0.001	1 000 000	-0.001	1 000 000
0.0001	1 000 000 000	-0.0001	1 000 000 000

It follows that the values of $f(x)$ do not approach a number and so, $\lim_{x\to 0}\frac{1}{x^2}$, does not exist.

To indicate such kind of behavior we write

$$\lim_{x\to 0}\frac{1}{x^2} = \infty$$

which neither regard ∞ as a number nor does it mean that the limit exists. It simply expresses the particular way in which the limit does not exist.

In general,

$$\lim_{x\to k} f(x) = \infty$$

indicates that the values of f(x) become larger and larger (or increase without bound) as $x \to k$. It often read as "the limit of $f(x)$, as x approaches k is infinity " or " $f(x)$ becomes infinite as x approaches k " or " $f(x)$ increases without bound as x approaches k ".

Similarly,

$$\lim_{x\to k} f(x) = -\infty$$

means that the values of $f(x)$ are as large negative as we like for all values of x that are sufficiently close to k , but not equal to k.

Example 1.5.8 (a) Evaluate $\lim_{x\to 0}\frac{1}{x^2+x^3}$ (b) $\lim_{x\to 0}-\frac{1}{x+x^2}$

Solution

(a) We divide numerator and denominator by x^2 to get

$$\lim_{x\to 0}\frac{1}{x^2+x^3}=\lim_{x\to 0}\frac{\frac{1}{x^2}}{1+x}=\frac{\lim_{x\to 0}\frac{1}{x^2}}{\lim_{x\to 0}(1+x)}$$

Now, $\lim_{x\to 0}\frac{1}{x^2}=\infty$, and $\lim_{x\to 0}(1+x)=1$

The numerator grows without bound, while the denominator approaches 1 implying that $\frac{1}{x^2+x^3}$

tends to infinity. Hence $\lim_{x\to 0}\frac{1}{x^2+x^3}=\infty$

(b) $\lim_{x\to 0}\frac{-1}{(x+x^2)}=\lim_{x\to 0}\frac{-\frac{1}{x}}{1+x}=\frac{\lim_{x\to 0}\left(-\frac{1}{x}\right)}{\lim_{x\to 0}(1+x)}$

But $\lim_{x\to 0}\left(-\frac{1}{x}\right)=\infty$ when $x<0$ and $\lim_{x\to 0}\left(-\frac{1}{x}\right)=-\infty$ if $x<0$ while $\lim_{x\to 0}(1+x)=1$

Thus the behavior of $\frac{-1}{(x+x^2)}$ depends on the direction from which x approaches zero. Hence

$\lim_{x\to 0}\frac{-1}{x+x^2}$ does not exists.

1.5.4.2 Limits at infinity

Recall that in computing infinite limits, we allow x to approach a number and the values of $y=f(x)$ became arbitrarily large (positive or negative). In this case, we let x become arbitrarily large (positive or negative) and see what happens to (x) .

Consider the function $f(x)=\frac{x^2-1}{x^2+1}$ as x becomes large. As presented in Table 2, if x grows larger and larger you can see that the values of $f(x)$ get closer and closer to 1. Indeed, we can make the values of $f(x)$ as close as we like to 1 by taking x sufficiently large. This situation is written symbolically by

$$\lim_{x\to\infty}\frac{x^2-1}{x^2+1}=1.$$

Table 3

x	$f(x)=\frac{x^2-1}{x^2+1}$	x	$f(x)=\frac{x^2-1}{x^2+1}$
1	0	-1	0
2	0.600000	-2	0.600000
3	0.800000	-3	0.800000
4	0.882353	-4	0.882353
5	0.923077	-5	0.923077
10	0.980198	-10	0.980198
50	0.999200	-50	0.999200
100	0.999800	-100	0.999800
1000	0.999998	-1000	0.999998

In general, $\lim_{x\to\infty} f(x)=L.$ indicates that the values of $f(x)$ approach L as x becomes larger and larger.

Definition 1.5.4.1 If $f(x)$ is a function defined on some interval $(a,\ \infty)$, then

$$\lim_{x\to\infty} f(x)=L.$$

This means that the values of $f(x)$ can be made as close to L as we like by taking x sufficiently large.

Note that $\lim_{x\to\infty} f(x)=L$ is alternatively expressed as $f(x)\to L$ as $x\to\infty$ and is often read as "the limit of (x) , as x approaches infinity, is L."

In a similar note,

$$\lim_{x\to-\infty} f(x)=L$$

indicates that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large negative.

Remark 1.5.4.1

(1) Observe that when x is large, $\frac{1}{x}$ is small and so, by taking x large enough, we can make $\frac{1}{x}$ as close to 0 as we like. Therefore, according to Definition 1.5.4.1, we get

$$\lim_{x\to\infty}\frac{1}{x}=0.$$

Similarly, when x is large negative, $\frac{1}{x}$ is small negative and so, also

$$\lim_{x\to-\infty}\frac{1}{x}=0.$$

(2) The limit laws presented in section 1.5.1 with exception of law 8 are also valid if "$x\to k$" is replaced by "$x\to\infty$" or "$x\to-\infty$".

(3) Combining law 6 with results of Remark (1) above, we have that if n is a positive integer, then

$$\lim_{x\to\infty}\frac{1}{x^n}=0 \text{ and } \lim_{x\to-\infty}\frac{1}{x^n}=0.$$

(4) To evaluate the limit at infinity of any rational function with degree of denominator greater or equal to degree of the numerator, first divide both the numerator and denominator by the highest power of x that occurs in the denominator.

Example 1.5.4.1 Evaluate (i) $\lim_{x\to\infty}\frac{3x+1}{x}$ (ii) $\lim_{x\to\infty}\frac{3x^2-x-2}{5x^2+4x+1}$

Solution

(i) We divide the numerator and denominator by x to obtain

$\lim_{x\to\infty}\frac{3x+1}{x}=\lim_{x\to\infty}\left(3+\frac{1}{x}\right)=\lim_{x\to\infty}3+\lim_{x\to\infty}\frac{1}{x}=3$ by Remark 1.5.4.1(3)

(ii) We divide numerator and denominator by x^2 to get

$$\lim_{x\to\infty}\frac{3x^2-x-2}{5x^{2+}4x+1}=\lim_{x\to\infty}\frac{3-\frac{1}{x}-\frac{2}{x^2}}{5+\frac{4}{x}+\frac{1}{x^2}}$$

$$=\frac{\lim_{x\to\infty}3-\lim_{x\to\infty}\frac{1}{x}-2\lim_{x\to\infty}\frac{1}{x^2}}{\lim_{x\to\infty}5+4\lim_{x\to\infty}\frac{1}{x}+\lim_{x\to\infty}\frac{1}{x^2}}=\frac{3}{5}$$

1.5.4.3 Infinite limits at infinity

We should note that

$$\lim_{x\to\infty}f(x)=\infty$$

indicates that the values of $f(x)$ become large as x becomes large. Similarly for the following

$$\lim_{x\to-\infty}f(x)=\infty,\ \lim_{x\to\infty}f(x)=-\infty \text{ and } \lim_{x\to-\infty}f(x)=-\infty$$

To evaluate the limit at infinity of any rational function with degree of the denominator less than the degree of the numerator, first divide both the numerator and denominator by the highest power of x that appears in the denominator.

Example 1.5.4.3 Evaluate (i) $\lim_{x\to\infty}\frac{x^2+x}{3-x}$ (ii) $\lim_{x\to-\infty}\frac{-4x^3+7x}{2x^2-3x-10}$

Solution

(i) $\lim_{x\to\infty}\frac{x^2+x}{3-x}=\lim_{x\to\infty}\frac{x+\frac{1}{x}}{\frac{3}{x}-1}=\frac{\lim_{x\to\infty}x+\lim_{x\to\infty}\frac{1}{x}}{3\lim_{x\to\infty}\frac{1}{x}-\lim_{x\to\infty}1}=-\infty$

(ii) $\lim_{x\to-\infty}\frac{-4x^3+7x}{2x^2-3x-10}=\lim_{x\to-\infty}\frac{-4x+\frac{7}{x}}{2-\frac{3}{x}-\frac{10}{x^2}}$

$$=\frac{-4\lim_{x\to-\infty}x+7\lim_{x\to-\infty}\frac{1}{x}}{\lim_{x\to-\infty}2-3\lim_{x\to-\infty}\left(-\frac{1}{x}\right)-10\lim_{x\to-\infty}\frac{1}{x^2}}$$

$$=\frac{-4\lim_{x\to-\infty}x}{\lim_{x\to-\infty}2}=\infty$$

Exercise

(1) Find the limit, if it exists (i) $\lim_{x\to4}\frac{x^2-4x}{x^2-3x-4}$ (ii) $\lim_{x\to-2}\frac{x+2}{x^3+8}$ (iii) $\lim_{x\to3}\frac{\sqrt{4u+1}-3}{u-2}$

(iv) $\lim_{x\to3}(x^4-3x)(x^2+5x+3)$ (v) $\lim_{x\to-2}\sqrt{u^4+3u+6}$

(2) If $\lim_{x\to1}\frac{f(x)-8}{x-1}=10$, find $\lim_{x\to1}f(x)$

(3) Evaluate (a) $\lim_{x\to\infty}\frac{x^2-3x+5}{x^2+2x-4}$ (b) $\lim_{x\to\infty}\frac{6x^5-3x^2+11x+1}{2x^5+5x^2+7}$ (c) $\lim_{x\to\infty}\frac{11x+2}{2x^2-1}$

(4) Find the limit (i) $\lim_{x\to0}\frac{sin4x}{sin6x}$ (ii) $\lim_{x\to0}\frac{sin3x}{5x^3-4x}$ (iii) $\lim_{x\to0}\frac{cosx-1}{sinx}$

(5) Let $g(x)=\frac{x^2+x-6}{|x-2|}$ (a) Find (i) $\lim_{x\to2^+}g(x)$ (ii) $\lim_{x\to2^-}g(x)$ (b) Does $\lim_{x\to2}g(x)$ exist?

1.5.5 Continuity of a function

We have seen that the limit of a function as x approaches k can often be found simply by calculating the value of the function at *k*. Functions with such a property are said to be continuous at k.

Definition 1.5.5.1 A function $f(x)$ is said to be continuous at a number x_0 if

$$\lim_{x\to x_0}f(x)=f(x_0).$$

It follows from this definition that a continuous function at a number satisfies the following three conditions.

(i) $f(x_0)$ exists (i.e., x_0 lies in the domain of *f*)

(ii) $\lim_{x\to x_0}f(x)$ exists (*f* has a limit as $x\to x_0$)

(iii) $f(x_0)=\lim_{x\to x_0}f(x)$ (the function value equals the limit)

A function $f(x)$ is said to be discontinuous at a number x_0, if it is defined near x_0 but not continuous at x_0. Physical phenomena are usually continuous. For example, displacement or velocity of a vehicle varies continuously with time. Hence the function $f(x) = x^2$ is continuous.

1.5.5.1 Types of discontinuity

A discontinuous function may have one of the following kind of discontinuity.

1. Infinite discontinuity: A function $f(x)$ has a jump discontinuity at a number x_0 when its limit does not exist at x_0 and there is no way to transform $f(x)$ to change the situation. Moreover, if one sided limits of $f(x)$ exist at x_0 with different values ($\lim_{x\to x_0^+} f(x)$ and $\lim_{x\to x_0^-} f(x)$ both exist with $\lim_{x\to x_0^+} f(x) \neq \lim_{x\to x_0^-} f(x)$), then $f(x)$ is said to have a jump discontinuity. For instance,

   (i) $f(x) = \frac{1}{x-2}$ has an infinite discontinuity at $x = 2$ since $f(x)$ is not defined at $x = 2$ and $\lim_{x\to 2} \frac{1}{x-2}$ does not exist.

   (ii) $f(x) = \frac{|x|}{x}$ has a jump discontinuity at $x = 0$ as $f(0)$ is not defined and

   $$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{|x|}{x} = \lim_{x\to 0} \frac{x}{x} = 1 \text{ and } \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{|x|}{x} = \lim_{x\to 0} \frac{-x}{x} = -1$$

   Thus $\lim_{x\to 0^-} \neq \lim_{x\to 0^+} f(x)$

2. Removable discontinuity: A function $f(x)$ is said to have a removable discontinuity at x_0 if $f(x)$ is not define at x_0 but it has a limit there.

   Example The function $f(x) = \frac{x^2-4}{x-2}$ has infinite discontinuity at $x = 2$ since f is not define at $x = 2$ but $\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x+2)(x-2)}{x-2} = \lim_{x\to 2}(x+2) = 4$. Then let $g(x) = x + 2$ be the extended function and so, $g(2) = 4$. Thus $f(x)$ has a removable discontinuity at $x = 2$

Definition 1.5.5.1 A function $f(x)$ is continuous from the right (left) at a number x_0 if

$$\lim_{x\to x_0^+} f(x) = f(x_0) \quad (\lim_{x\to x_0^-} f(x) = f(x_0))$$

Definition 1.5.5.2 A function $f(x)$ is continuous

(i) on an interval I if it is continuous at every point in the interval,

(ii) at an endpoint of I *if* $f(x)$ is defined only on one side of the endpoint. Continuous at the endpoint means continuous from the right or continuous from the left.

For example, the function $f(x) = 1 - \sqrt{1-x^2}$ is continuous on the interval $[-1, 1]$ as if $-1 < x_0 < 1$, then $\lim_{x \to x_0} \left(1 - \sqrt{1-x^2}\right) = 1 - \sqrt{1-x_0^2} = f(x_0)$

Thus, $f(x)$ is continuous at x_0 if $-1 < x_0 < 1$.

Similarly,

$$\lim_{x \to -1^+} \left(1 - \sqrt{1-x^2}\right) = 1 = f(-1) \text{ and } \lim_{x \to 1^-} \left(1 - \sqrt{1-x^2}\right) = 1 = f(1)$$

Therefore, $f(x)$ is continuous from the right of -1 and continuous from the left of 1. Hence $f(x)$ is continuous on $[-1, 1]$.

1.5.5.2 Properties of continuity

Theorem 1.5.5.1 Suppose $f(x)$ and $g(x)$ are continuous at x_o and c is a constant, then

(a) $cf(x)$ is continuous at x_o for any constant c.

(b) $f(x) + g(x)$ is continuous at x_o.

(c) $f(x) - g(x)$ is continuous at x_o.

(d) $f(x)g(x)$ is continuous at x_o.

(e) $\frac{f(x)}{g(x)}$ continuous at x_o, if $g(x_0) \neq 0$

Corollary 1.5.5.1 if $f(x)$ and $g(x)$ are continuous on an interval, then so are the functions $cf, f+g, f-g, f.g$ and $\frac{f}{g}$ if $g \neq 0$.

Theorem 1.5.5.2 The following types of functions: polynomials, rational functions, root functions, and trigonometric functions are continuous at every number in their domains.

For example, $f(x) = x^5 + 3x^2 - 45, g(x) = \frac{x-7}{x^2+1}, h(x) = \sqrt{x}$ are continuous on $(-\infty, \infty), (-\infty, \infty)$ and $[0, \infty)$, respectively.

Theorem 1.5.5.3 Suppose $g(x)$ is continuous at x_0 and $f(x)$ is continuous at $g(x_0)$, then the composite function $f \circ g$, denoted by $(f \circ g)(x) = f(g(x))$ is continuous at x_0. For instance, $h(x) = \sin(x^2)$ is continuous on $\mathbb{R}$ since $g(x) = x^2$ which is continuous on $\mathbb{R}$ and $f(x) = \sin x$ which is continuous everywhere.

Theorem 1.5.5.4 (The intermediate value theorem)

If $y = f(x)$ is continuous on a closed interval $[a, b]$ and y_0 any number between $f(a)$ and $f(b)$ with $f(a) \neq f(b)$, then there exists a number c in (a, b) such that $f(c) = y_0$.

The intermediate value theorem states that a function $y= f(x)$ that is continuous on a closed interval $[a, b]$ takes on every value between $f(a)$ and $f(b)$. That is. if y_o is any value between $f(a)$ and $f(b)$, then $f(c) = y_o$ for some c in (a, b).

One application of the Intermediate Value Theorem is in locating roots of equations as in the following example.

Example Show that there is a root of the equation $4x^3 - 6x^2 + 3x - 2 = 0$ between 1 and 2.

Solution

Given $f(x) = 4x^3 - 6x^2 + 3x - 2$. we are to look for the solution of this equation. That is, a number c between 1 and 2 such that $f(c) = 0$. Therefore, taking $a = 1,\ b = 2$ and $y_0 = 0$, we have

$$f(1) = 1 - 6 + 3 - 2 = -4 < 0 \text{ and } f(2) = 4 \times 2^3 - 6 \times 2^2 + 3 \times 2 - 2 = 12 > 0.$$

It follows that, $f(1) < 0 < f(2)$ and so, $y_0 = 0$ is a number between $f(1)$ and $f(2)$. Thus, the equation $f(x)$ has at least one root c in the interval $(1, 2)$.

To locate the root, we apply the Intermediate value theorem again. Now,

$$f(1.2) = -0.128 < 0 \text{ and } f(1.3) = 0.548 > 0$$

Implying that a root must lie between 1.2 and 1.3. Then

$$f(1.22) = -0.007008 < 0 \text{ and } f(1.23) = 0.056068 > 0$$

So that a root lies in the interval (1.22, 1.23**).**

Exercise

1. Use the definition of continuity and the properties of limits to show that the function is continuous at the given number a
   (i) $f(x) = 3x^4 - 5x + \sqrt[3]{x^2 + 4}, \quad a = 2$
   (ii) $f(x) = (x + 2x^3)^4, \quad a = -1$
2. Use the definition of continuity and the properties of limits to show that the function $f(x) = x\sqrt{16 - x^2}, \quad a = -1$ is continuous on the interval $[-4, 4]$.
3. Explain why the function is discontinuous at the given number .
   a. $f(x) = \frac{1}{x+2}, \quad x = -2$ (b) $f(x) = \begin{cases} \frac{x^2-x}{x^2-1} & if\ x \neq 1 \\ 1, & if\ x = 1 \end{cases} \quad a = 1$
4. Explain why the function is continuous at every number in its domain. State the domain. (i) $f(x) = \frac{2x^2-x-1}{x^2+1}$ (ii) $g(x) = \sqrt{1 + \frac{1}{x}}$

5. If a and b are positive numbers, prove that the equation $\frac{a}{x^3+2x^2-1}+\frac{b}{x^3+x-2}=0$ has at least one solution on the interval $(-1,1)$.

6. Test the continuity or otherwise of $f(x)=\dfrac{\sin x}{x}$ for $x \neq 0$.

SECTION B

Differentiation and its applications

Definition 2.1 The derivative of a function f at a number h, denoted by $f'(x)$, is

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

if this limit exists.

2.1 Basic differentiation formulas

2.1.1 Derivative of constant function

Consider the function $f(x)=c$, then $f'(x)=0$.

Proof

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{c-c}{h}=\lim_{h\to 0}0=0$$

For example, the derivatives of $f(x)=\pi^2, g(x)=-\frac{\sqrt{3}}{2}$ are respectively, $f'(x)=0, g'(x)=0$.

2.1.2 Power functions

If n is a positive integer and $f(x)=x^n$, then $f'(x)=nx^{n-1}$.

Proof

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{(x+h)^n-x^n}{h}$$

Applying the expansion of $(x+h)^n$ by using Binomial theorem, we have

$$f'(x)=\lim_{h\to 0}\frac{\left\{x^n+nx^{n-1}h+\frac{n(n-1)}{2}x^{n-2}h^2+\cdots+nxh^{n-1}+h^{n-1}\right\}-x^n}{h}$$

Simplifying this, gives

$$f'(x)=nx^{n-1}$$

Note that the power rule is true for any real number n.

Example 2.1.2.1

(a) If $f(x) = x^6$ then $f'(x) = 6x^5$

(b) If $f(x) = 2x^{10}$, then $f'(x) = 20x^9$

(c) If $f(x) = \sqrt{x}$, then $f'(x) = \frac{1}{2\sqrt{x}}$

Exercise 2.1.2.1 Find $f'(x)$ if (a) $f(x) = x^4$ (b) $f(x) = \frac{1}{x^2}$ (c) $f(x) = \sqrt[3]{x^2}$

2.1.3 Constant multiple rules

Theorem 2.1.3.1 If *f* is a differentiable function and *k* a constant, then

$$\frac{d(kf)}{dx} = k\frac{df}{dx}$$

Proof

$$\frac{d(kf)}{dx} = \lim_{h\to 0}\frac{kf(x+h) - kf(x)}{h} = kf'(x)$$

Examples 2.1.3.1

Evaluate $\frac{dy}{dx}$ (a) $y = 6x^3$ (b) $y = \frac{3}{5}x^{\frac{6}{5}}$

Solution: (a) $\frac{dy}{dx} = 6\times 3\times x^{3-1} = 18x^2$ (b) $\frac{dy}{dx} = \frac{3}{5}\times\frac{6}{5}x^{\frac{6}{5}-1} = \frac{18}{25}x^{\frac{1}{5}}$

2.1.4 Derivatives of sum and difference of two functions

Theorem 2.1.4.1 If *f* and *g* are differentiable functions, then so is *f* + *g* and *f* - *g*. Their derivatives are given as:

$$\frac{d}{dx}(f+g) = (f+g)' = \frac{df}{dx} + \frac{dg}{dx} = f' + g' \quad (1)$$

$$\frac{d}{dx}(f-g) = (f-g)' = \frac{df}{dx} - \frac{dg}{dx} = f' - g' \quad (2)$$

Proof

$$\frac{d}{dx}[f(x)+g(x)] = \lim_{h\to 0}\frac{[f(x+h)+g(x+h)]-[f(x)+g(x)]}{h}$$

$$= \lim_{x\to 0}\frac{f(x+h)-f(x)}{h} + \lim_{h\to 0}\frac{g(x+h)-g(x)}{h}$$

$$= \frac{df}{dx} + \frac{dg}{dx}$$

Examples 2.1.4.1

Differentiate (a) $5x^2 + 2x^4$ (b) $3x^2 - \sqrt{x} + 3$

Solution

(a) $\frac{d}{dx}(5x^2)+\frac{d}{dx}(2x^4)=10x+8x^3$

(b) $\frac{d}{dx}(3x^2-\sqrt{x}+3)=\frac{d}{dx}(3x^3)-\frac{d}{dx}(\sqrt{x})+\frac{d}{dx}(3)=9x^2-\frac{1}{2\sqrt{x}}$

Exercise 2.1.4.1

Find the derivative if (a) $y = x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5$ (b) $y = x^4 - 6x^2 + 4$

2.2 Higher order derivatives

$\frac{dy}{dx} = y'$ is called the first (first order) derivative of y with respect to x. This derivative may itself be a differentiable function of *x*, if so, its second derivative is

$$y'' = \frac{dy'}{dx} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

Similarly, the third derivative is $y''' = \frac{dy''}{dx} = \frac{d^3y}{dx^3}$.

In general, for any positive integer n, the nth order derivative of y with respect to x is

$$y^n = \frac{d}{dx}\left(y^{n-1}\right).$$

Examples 2.2.1

Given $y = x^5 - 3x^4 + 2x^2 + 5$. Compute $y', y'', y''', y^{iv}, y^{v}, y^{vi}$

Solution:

$y' = 5x^4 - 12x^3 + 4x$, $y'' = 20x^3 - 36x + 4$, $y''' = 60x^2 - 36$, $y^{iv} = 120x$, $y^{v} = 120$, $y^{vi} = 0$

2.3 Differentiation Techniques

2.3.1 Product rule

Theorem 2.3.1.1 If *f* and *g* are differentiable at *x*, then so is their product, and

$$\frac{d}{dx}(fg) = f\frac{dg}{dx} + g\frac{df}{dx} \Rightarrow \left[(fg)' = f'g + g'f\right]$$

Proof

$$\frac{d}{dx}(fg) = \lim_{h\to 0}\frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

To change this fraction into an equivalent one that contains difference quotients for the derivatives of *f* and *g*, we add and subtract $f(x + h)g(x)$ in the numerator.

$$\frac{d}{dx}(fg) = \left[\lim_{h\to 0}\frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}\right]$$

$$= \lim_{h\to 0}\left[\frac{f(x+h)g(x+h)-g(x)}{h}+\frac{g(x)f(x+h)-f(x)}{h}\right]$$

$$= \lim_{h\to 0} f(x+h)\cdot\lim_{h\to 0}\frac{g(x+h)-g(x)}{h}+g(x)\cdot\lim_{h\to 0}\frac{f(x+h)f(x)}{h}$$

$$\frac{d}{dx}(fg)= f\frac{dg}{dx}+g\frac{df}{dx}$$

Example 2.3.1.1 If (a) $y=(x^2-3)(x+1)^2$ (b) $y=(3x^2-5x^3)(x^5-\sqrt{x}+2)$

find $\frac{dy}{dx}$ and simplify the result.

Solution

(a) Let $f=x^2-3$ and $g=(x+1)^2$, then $\frac{df}{dx}=2x$, $\frac{dg}{dx}=2(x+1)$

$$\frac{dy}{dx}=(x+1)^2.2x+(x^2-3).2(x+1)$$
$$=2(x+1)[x(x+1)+x^2-3]$$
$$=2(x+1)(2x^2+x-3)$$
$$\frac{dy}{dx}=2(x+1)(2x^2-2x+3x-3)$$
$$=2(x+1)(2x(x-1)+3(x-1)$$
$$=2(x+1)(x-1)(2x+3)$$

(b) $\frac{dy}{dx}=(3x^2-5x^3)(x^5-\sqrt{x}+2)$

$$=(3x^2-5x^3)\frac{d}{dx}(x^5-\sqrt{x}+2)+(x^5-\sqrt{x}+2)\frac{d}{dx}(3x^2-5x^3)$$
$$=(3x^2-5x^3)\left(5x^4-\frac{1}{2\sqrt{x}}\right)+(x^5-\sqrt{x}+2)(6x-15x^2)$$

Exercise 2.3.1.1

(1) Find $\frac{dy}{dx}$ if $y=\frac{x}{x+1}$

(2) Use product rule to find the derivative of *y* w.r.t. *x* of $y=\frac{1}{x}\left(x^2+\frac{1}{x}\right)$

2.3.2 Quotient rule

Theorem 2.3.2.1 If *f* and *g* are differentiable function of *x* and $g(x)\neq 0$, then the quotient $\frac{f}{g}$ is differentiable at *x* and

$$\frac{d}{dx}\left(\frac{f}{g}\right)=\frac{g\dfrac{df}{dx}-f\dfrac{dg}{dx}}{g^2}\Rightarrow\left[\left(\frac{f}{g}\right)'=\frac{gf'-fg'}{g^2}\right]$$

Proof

$$\frac{d}{dx}\left(\frac{f}{g}\right)=\lim_{h\to 0}\left[\frac{\frac{f(x+h)}{g(x+h)}-\frac{f(x)}{g(x)}}{h}\right]$$

$$=\lim_{h\to 0}\frac{f(x+h)g(x)-f(x)g(x+h)}{hg(x+h)g(x)}$$

We subtract and add $g(x)\,f(x)$ in the numerator, we then have

$$\frac{d}{dx}\left(\frac{f}{g}\right)=\lim_{h\to o}\frac{\left[f(x+h)g(x)-g(x)f(x)+g(x)f(x)-f(x)g(x+h)\right]}{hg(x+h)f(x)}$$

$$=\lim_{h\to 0}\left[\frac{g(x)\frac{f(x+h)-f(x)}{h}-f(x)\frac{g(x+h)-g(x)}{h}}{g(x+h)g(x)}\right]$$

Therefore

$$\frac{d}{dx}\left(\frac{f}{g}\right)=\frac{g\frac{df}{dx}-f\frac{dg}{dx}}{g^2}$$

Example2.3.2.1 Use quotient rule to find $\dfrac{dy}{dx}$ given that $f(x)=\dfrac{2x^3}{x^2-4}$

Solution

Let $f=2x^3$, $\mathrm{g}=x^2-4$, then

$$\frac{dy}{dx}=\frac{\left(x^2-4\right)\dfrac{d}{dx}\left(2x^3\right)-2x^3\dfrac{d}{dx}\left(x^2-4\right)}{\left(x^2-4\right)^2}$$

$$=\frac{6x^2(x^2-4)-4x^4}{(x^2-4)^2}$$

$$=\frac{2x^2(x^2-12)}{(x^2-4)^2}$$

2.3.3 Composite function

If $y=f(u)$ and $u=g(x)$ are both differentiable functions, the

$$\frac{dy}{dx}=\frac{dy}{du}\times\frac{du}{dx}$$

Example2.3.3.1

Find $\frac{dy}{dx}$, if $y = 8x - 12 = 4(2x - 6)$

Solution

$y = 4u$, where $u = 2x - 6$ and $\frac{dy}{dx} = 8$, $\frac{dy}{du} = 4$, $\frac{du}{dx} = 2$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 8$$

Example2.3.3.2 Differentiate $\left(x^2 + 3x\right)^6$

Solution

Let $y = \left(x^2 + 3x\right)^6$, $u = x^2 + 3x$, then $y = u^6$ and $\frac{du}{dy} = 6u^5$, $\frac{du}{dx} = 2x + 3$

Therefore, $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = 6u^5(2x + 3) = 6(x^2 + 3x)^5(2x + 3)$

Example2.3.3.3 Differentiate $\frac{1}{1+\sqrt{x}}$

Solution: Let $y = \frac{1}{1+\sqrt{x}} = \left(1+\sqrt{x}\right)^{-1}$, then $\frac{dy}{dx} = -\left(\frac{1}{1+\sqrt{x}}\right)^{-2} \frac{1}{2} x^{-\frac{1}{2}} = \frac{-1}{2\sqrt{x}\left(\frac{1}{1+\sqrt{x}}\right)^2}$

2.4 Derivatives of trigonometric function

2.4.1 Derivative of sin x

Let $y = \sin x$, then $\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h\to 0} \frac{\sin x \cosh + \sinh \cos x - \sin x}{h}$$

Using $\sin(x+h) = \sin x \cosh + \sinh \cos x$, we have

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{\sin x \cosh - \sin x}{h} - \lim_{h\to 0} \frac{\cos x \sinh}{h}$$

$$= \lim_{h\to 0} \frac{\sin x(\cos x - 1)}{h} + \lim_{h\to 0} \frac{\sinh}{h}(\cos x)$$

$$= \sin x \cdot 0 + 1 \cdot \cos x = \cos x$$

In general, $\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$, where u is a function of x.

Example 2.4.1.1 Differentiate (a) $y = x^2 - \sin x$ (b) $y = x^3 \sin x$ (c) $y = 3\sin^4 2x$

Solution

(a) $\frac{dy}{dx} = \frac{d}{dx}(x^2) - \frac{d}{dx}(\sin x) = 2x - \sin x$

(b) $\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx} = \sin x \frac{d}{dx}x^3 + x^3 \frac{d}{dx}\sin x = 3x^2 \sin x + x^3 \cos x$

(c) $\frac{d}{dx}(3\sin^4 2x) = 3 \cdot 4\sin^3 2x \frac{d}{dx}(\sin 2x) = 24\sin^3 2x \cos 2x$

2.4.2 Derivative of Cos *x*

If $y = \cos x$ then using $\cos(x+h) = \cos x \cosh - \sinh \sin x$, we have

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0}\frac{\cos(x+h) - \cos x}{h}$$

$$= \lim_{h\to 0}\frac{\cos x \cosh - \sin x \sinh - \cos x}{h}$$

$$= \lim_{h\to 0}\frac{\cos x \cosh - \cos x}{h} - \lim_{h\to 0}\frac{\sin x \sinh}{h}$$

$$= \lim_{h\to 0}\frac{\cos x(\cosh - 1)}{h} - \lim_{h\to 0}\frac{\sin x \sinh}{h}$$

$$= \cos x \lim_{h\to 0}\frac{\cosh - 1}{h} - \sin x \lim_{h\to 0}\frac{\sinh}{h}$$

$$= \cos x \cdot 0 - \sin x \cdot 1$$

$$= -\sin x$$

In general, $\frac{d}{dx}(\cos u) = -\sin u \frac{du}{dx}$, where *u* is a function of *x*.

Example2.4.2.1 Find $\frac{dy}{dx}$, if (a) $y = \sin x \cos x$ (b) $y = \frac{\cos x}{1 - \sin x}$

Solution

(a) $\frac{dy}{dx} = \cos x \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(\cos x) = cos^2 x - sin^2 x = \sin 2x$

(b) $\frac{dy}{dx} = \frac{(1-\sin x)(-\sin x) - \cos x(-\cos x)}{(1-\sin x)^2} = \frac{-\sin x + sin^2 x + cos^2 x}{(1-\sin x)^2} = \frac{1-\sin x}{(1-\sin x)^2} = \frac{1}{1-\sin x}$

2.4.3 Derivative of tangent function

Given $y = \tan x = \dfrac{\sin x}{\cos x}$,

Applying the Quotient rule, we obtain

$$\frac{dy}{dx} = \frac{u\dfrac{dv}{dx} - v\dfrac{du}{dx}}{v^2} = \frac{\sin x \dfrac{d}{dx}(\cos x) - \cos x \dfrac{d}{dx}(\sin x)}{\cos^2 x}$$

$$= \frac{\sin x(-\sin x) - \cos x \cos x}{\cos^2 x}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\cos^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\cos^2 x} = \frac{-1}{\cos^2 x} = -\sec^2 x$$

In general, $\dfrac{d}{dx}(\tan u) = -\sec^2 u \dfrac{du}{dx}$, where u is a function of x.

Moreover, if u is a function of x, then

$$\frac{d}{dx}(\sec u) = \sec u \tan u \frac{du}{dx}, \quad \left[\sec u = \frac{1}{\cos u}\right]$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u, \quad \left[\cot u = \frac{\cos u}{\sin u}\right]$$

$$\frac{d}{dx}(\csc u) = -\csc u \cot u, \quad \left[\csc u = \frac{1}{\sin u}\right]$$

Example 2.4.3.1 Find $\dfrac{d^2y}{dx^2}$, if $y = \sec x$

Solution

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d^2x}{dx^2} = \tan x \frac{d}{dx}(\sec x) + \sec x \frac{d}{dx}(\tan x) = \tan^2 x \sec x + \sec^3 x$$

Example 2.4.3.2 Find the derivative of $g(t)$ = tan(5-sin2t)

Solution

$$g(t) = \tan(5 - \sin 2t)$$

$$g'(t) = \frac{dg}{dt} = \frac{d}{dt}(\tan(5 - \sin 2t)) = -2\sec^2(5 - \sin 2t)\cos 2t.$$

2.5 Implicit and parametric differentiation

2.5.1 First order implicit differentiation

A function $y = f(x)$ is a function described by expressing y explicitly in terms of x. Such a function is called an explicit function while a function defined implicitly by a relation between x and y is known as an explicit function. For example, $x^3y - 5x = 3$ and $x^2 + y^2 = 1$ are implicit functions.

Example 2.5.1.1 Find $\frac{dy}{dx}$, if $x^5 + 4xy - 3y^5 = 2$

Solution

Given $x^5 + 4xy - 3y^5 = 2$, we have

$$\frac{d}{dx}(x^5) + 4\frac{d}{dx}(xy^3) - 3\frac{d}{dx}(y^5) = \frac{d}{dx}(2)$$

$$5x^4 + 4y^3 + 12xy^2\frac{dy}{dx} - 15y^4\frac{dy}{dx} = 0$$

$$5x^4 + 4y^3 + 3y^2\frac{dy}{dx}(4x - 5y^2) = 0$$

$$3y^2\frac{dy}{dx}(4x - 5y^2) = -(5x^4 + 4y^3)$$

$$\frac{dy}{dx} = -\frac{5x^4 + 4y^3}{3y^2(4x - 5y^2)}$$

Example 2.5.1.2 Find $\frac{dy}{dx}$, if $y^2 = x^2 + \sin xy$

Solution

Given $y^2 = x^2 + \sin xy$, differentiating both sides w.r.t x.

$$\frac{d}{dx}(y^2) = \frac{d}{dx}\left(x^2\right) + \frac{d}{dx}\left(\sin xy\right)$$

$$2y\frac{dy}{dx} = 2x + \cos xy\frac{d}{dx}\left(xy\right)$$

$$2y\frac{dy}{dx} = 2x + \left(x\frac{dy}{dx} + y\right)\cos xy$$

$$(2y - x\cos xy)\frac{dy}{dx} = 2x + y\cos xy$$

$$\frac{dy}{dx} = \frac{2x + y\cos xy}{2y - x\cos xy}$$

2.5.2Implicit derivatives of higher order

The higher order derivatives of an implicit function can also be obtained in a similar way we get those for the explicit function.

Example 2.5.2.1 Find $\frac{d^2y}{dx^2}$, if $2x^3 - 3y^2 = 8$

Solution $2x^3 - 3y^2 = 8 \Rightarrow x^2 - y\frac{dy}{dx} = 0$ and so, $\frac{dy}{dx} = \frac{x^2}{y}$

To find the second derivative, we differentiate the derivative function $\frac{dy}{dx}$

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{x^2}{y}\right)$$

$$\frac{d^2y}{dx^2} = -x^2y^{-2}\frac{dy}{dx} + 2xy^{-1}$$

$$= -\frac{x^2}{y^2}\frac{dy}{dx} + \frac{2x}{y} = \frac{2x}{y} - \frac{x^4}{y^3}, y \neq o$$

2.5.3 Parametric differentiation

When both *x* and *y* are written as function of another variable, we call the new variable a parameter. For example, $x = 2t + 3, y = t^2 + ½$ is a parametric equation with parameter t.

Examples 2.5.3.1 Find $\frac{dy}{dx}$ in terms of *t* given that $x = t^3 + t^2, y = t^2 + t$

Solution Given $x = t^3 + t^2, y = t^2 + t \Rightarrow \frac{dx}{dt} = 3t^2 + 2t, \quad \frac{dy}{dt} = 2t + 1$

Using the chain rule, we get $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{2t+1}{3t^2+2t}$

Exercise 2.5.3.1

(1) Given $x = t + \frac{1}{t}$, $y = t - \frac{1}{t}$, find $\frac{dy}{dx}$.

(2) Find $\frac{d^2y}{dx^2}$ as function of *t*, if $x = t - t^2$, $y = t - t^3$

2.6 Derivative of the natural logarithm function

The derivative of $\ln x$ is given by

$$\frac{d}{dx}(\ln x) = \frac{1}{x}.$$

In general, given $y = \ln u$ where u is a differentiable function of x, we obtain by chain rule that

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{u} \cdot \frac{du}{dx}$$

Example 2.6.1 Find $\frac{dy}{dx}$, if (a) $y = \ln 2x$ (b) $y = \ln(3x^2 + 4)$ (c) $y = \ln\cos x$

Solution (a) $\frac{d}{dx}(\ln 2x) = \frac{1}{2x} \cdot \frac{d}{dx}(2x) = \frac{1}{2x} \times 2 = \frac{1}{x}$

(b) Given $y = \ln(3x^2 + 4)$,

$$\frac{d}{dx}\left(\ln(3x^2 + 4)\right) = \frac{1}{3x^2 + 4} \cdot \frac{d}{dx}\left(3x^2 + 4\right) = \frac{1}{3x^2 + 4} \cdot 6x = \frac{6x}{3x^2 + 4}$$

(c) Given $y = \ln\cos x \Rightarrow \frac{dy}{dx} = \frac{1}{\cos x} \cdot \frac{d}{dx}(\cos x) = \frac{-\sin x}{\cos x} = -\tan x$

Exercise 2.6.1

(a) Differentiate $y = \ln(x^3 + 1)$ (c) Differentiate $y = \sqrt{\ln x}$

(b) Find $\frac{d}{dx}\ln(\sin x)$ (d) Find $\frac{d}{dx}\ln\frac{x+1}{\sqrt{x-2}}$

2.7 Derivative of the Natural exponential function

$$\frac{d}{dx}e^x = e^x$$

In general, given $y = e^u$, where u a differentiable function of x by the chain rule the derivative is.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left(e^u\right) \cdot \frac{du}{dx} \Rightarrow \frac{dy}{dx} = e^u \frac{du}{dx}$$

$$\therefore \frac{d}{dx}\left(e^u\right) = e^u \frac{du}{dx}$$

Example 2.7.1 Give (a) $y = e^{3x^2}$ (b) $y = e^{\sin x}$ (c) $y = e^{\sin x^2}$, find $\frac{dy}{dx}$

Solution

(a) $\frac{dy}{dx} = e^{3x^2} \frac{d}{dx}\left(3x^2\right) = e^{3x^2} \times 6x = 6xe^{3x^2}$

(b) $\frac{dy}{dx} = e^{\sin x} \frac{dy}{dx}(\sin x) = e^{\sin x}\cos x = \cos x e^{\sin x}$

(c) $\frac{dy}{dx} = e^{\sin x^2} \frac{dt}{dx}\left(\sin x^2\right) = e^{\sin x^2}\cos x^2 \frac{d}{dx}\left(x^2\right) = 2xe^{\sin x^2}\cos x^2$

Exercise 2.7.1 Differentiate

(a) $y = e^{tanx}$ (b) $y = \sqrt{x}e^x$ (c) $y = e^{-4x}sin5x$ (d) $y = (x^3 + 2x)e^x$

2.8 The general exponential function a^u

The exponential function $f(x) = a^x, a > 0$, is differentiable and

$$\frac{d}{dx}(a^x) = a^x \ln a$$

In general, let $y = a^u$,

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx} = \frac{d}{dx}\left(a^u\right)\frac{du}{dx}$$

$$\Rightarrow a^u \ln a \frac{du}{dx}$$

$$\therefore \frac{d}{dx}\left(a^u\right) = a^u \ln a \frac{du}{dx}$$

Example 2.8.1 Given (a) $y = 3^x$ (b) $y = 3^{-x}$ (c) $y = 3^{\sin 3x^2}$, find $\frac{dy}{dx}$

Solution

(a) Given $y = 3^x$

$$\frac{d}{dx}\left(3^x\right) = 3^x \ln 3 \frac{d}{dx}(x) = 3^x \ln 3$$

(b) $\frac{dy}{dx} = \frac{d}{dx}\left(3^{-x}\right) = 3^{-x}\ln 3\frac{d}{dx}(-x) = -3^{-x}\ln 3$

(c) Given $y = 3^{\sin 3x^2}$

$$\frac{dy}{dx} = 3^{\sin 3x^2}\ln 3\frac{d}{dx}\left(\sin 3x^2\right) = 3^{\sin 3x^2}\ln 3\cos 3x^2\cdot 6x = 6x\left(3^{\sin 3x^2}\right)\ln 3$$

Exercise 2.8.1

Find y' if (a) $y = 10^{x^2}$ (b) $y = (\sin x)^{\ln x}$ (c) $y = 10^{1-x^2}$ (d) $y = \frac{\ln x}{x^2}$

2.9 Applications of Differentiation

2.9.1 Indeterminate forms and L'hopital's rule

If $f(x)$ and $g(x)$ both approach zero ($\pm\infty$) as x approaches a then $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$,

respectively which is indeterminate and to overcome this problem L'hopital's rule was introduced. That is, if $f(x)$ and $g(x)$ both approach zero or infinity as x approaches a, then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$$

Remark 2.9.1

1. L'Hospital's Rule is also valid for one-sided limits and for limits at infinity or negative infinity; that is, "$x \to a$" can be replaced by any of the symbols $x \to a^+, x \to a^-, x \to \infty,$ or $x \to -\infty$.

2. Read on the following respective indeterminate forms for product, differences and powers $0.\infty, \infty - \infty, 0^0, \infty^0$ and 1^∞

Example 2.9.1: Evaluate $\lim_{x\to 0}\frac{\sin x}{x}$ **Solution:** $\lim_{x\to 0}\frac{\sin x}{x} = \lim_{x\to 0}\frac{\cos x}{1} = 1$

Example 2.9.2: Evaluate $\lim_{x\to 0}\frac{\sqrt{1+x}-1-\frac{x}{2}}{x^2}$

Solution: $\lim_{x\to o}\frac{\sqrt{1+x}-1-\frac{x}{2}}{x^2} = \lim_{x\to 0}\frac{\frac{1}{2}(1+x)^{-1/2}-\frac{1}{2}}{2x} = \lim_{x\to 0}\frac{-\frac{1}{4}(1+x)^{-\frac{1}{2}}}{2} = -\frac{1}{8}$

Example 2.9.3: Evaluate $\lim_{x\to 0}\frac{1-\cos x}{x+x^2}$ **Solution:** $\lim_{x\to 0}\frac{1-\cos x}{x+x^2} = \lim_{x\to 0}\frac{\sin x}{1+2x} = 0$

Example 2.9.4: Evaluate $\lim_{x\to 0^+}\frac{\sin x}{x^2}$ **Solution:** $\lim_{x\to 0^+}\frac{\sin x}{x^2} = \lim_{x\to 0^+}\frac{\cos x}{2x} = \infty$

Exercise 2.9.1

Evaluate the following:

(a) Find $\lim_{x\to 1}\frac{\ln x}{x-1}$ (b) Find $\lim_{x\to\infty}\frac{e^x}{x^2}$ (c) $\lim_{x\to\infty}\frac{\ln x}{\sqrt[3]{x}}$ (d) $\lim_{x\to 0}\frac{tanx-x}{x^3}$

2.9.2 Maxima and Minima

Here we learn how derivatives help us locate maximum and minimum values of functions. This is pertinent as many practical problems require us to minimize a cost or maximize an area or somehow find the best possible outcome of a situation.

2.9.2.1 Absolute (Global) Extreme value

Absolute (or global) maximum and minimum values are called absolute extrema.

Definition 2.9.2 Let f be a function with domain D and c be a number in D. Then $f(c)$ is the

(a) absolute (global) maximum value on D if and only if $f(x) \le f(c)$ for all x in D.

(b) absolute (global) minimum value on D if and only if $f(x) \ge f(c)$ for all x in D.

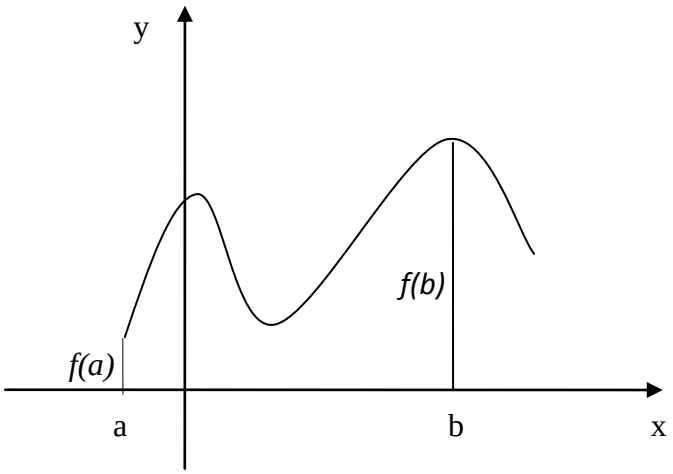


Figure 3. Absolute min. *f(a)* and Absolute max. *f(b)*

Theorem 2.9.2 (Extreme value theorem for continuous function)

If $f(x)$ is continuous at every point of a closed interval I, then $f(x)$ assumes both an absolute maximum M and an absolute minimum value m somewhere in I. That is, there are numbers x_1, and x_2 in I with $f(x_1) = m, f(x_2) = M$.

2.9.3 Possible maximum and minimum on a closed interval [a, b]

(i) maximum and minimum at interior points.

(ii) maximum and minimum at endpoints

(iii) Maximum at interior point and minimum at endpoint.

(iv) Minimum at interior point and maximum at end point.

2.9.4 Local Extreme values

Local extrema are also called relative extrema.

Definition 2.9.2 Let c be an interior point of the domain of a function $f(x)$, then $f(c)$ is a

a. Local maximum value at c if $f(x) \leq f(c)$ for all x in some open interval containing c.

b. Local minimum value at c if $f(x) \geq f(c)$ for all x in some open interval containing c.

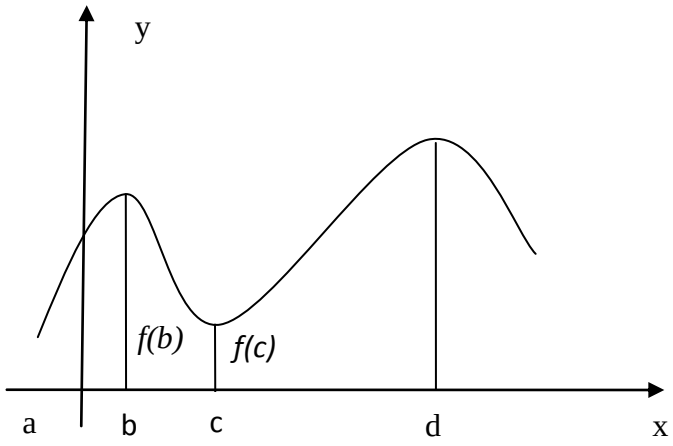


Figure 4. Local max. *f(b)* and Local min. *f(c)*

It is shown in Figure 4 that $f(b)$ is the local maximum value of f considering the interval (a, c). While $f(c)$ is the local minimum value of f in the interval (b, d).

Theorem 2.9.2 (Fermat's theorem)

If a function $f(x)$ has a local maximum value or a local minimum value at an interior point c of its domain and if $f'(x)$ exists at c, then $f'(c) = 0$.

Note that when the derivative of f is zero at c, f does not necessarily have a maximum or minimum at c. That is, the converse of the Fermat's Theorem is false in general.

2.9.5 Critical point

Definition 2.9.3 A point in the domain of a function $f(x)$ at which $f'(x) = 0$ or $f'(x)$ does not exist is called a critical point of $f(x)$.

Remark 3.2.4

The Fermat's Theorem can be expressed in terms of critical point as if f has a local maximum or local minimum at c, then c is a critical point of f.

To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$, we need to follow the three-step procedure below:

(i) Find the values of f at the critical numbers of f in (a, b).

(ii) Find the values of f at the endpoints of the interval.

(iii) The largest of these values (i) and (ii) is the absolute maximum value of f. While the smallest of these values is the absolute minimum value of f.

Example 2.9.1

Find the absolute maximum and minimum values of $f(x)=x^3-3x^2+1$ on the interval $\left[-\frac{1}{2},4\right]$.

Solution

We evaluate the function $f(x)$ at the critical points and endpoints. Then, we take the largest and the smallest of these values.

Now, $f'(x)=3x^2-6x=3x(x-2)$ have critical points at $x=0, x=2$ with values $f(0)=1$, $f(2)=-3$. These critical numbers are in the interval $\left(-\frac{1}{2},4\right)$.

At the endpoints, the values of f are: $f\left(-\frac{1}{2}\right)=\frac{1}{8}$ and $f(4)=17$.

Comparing the four values of f, the absolute minimum value is -3, and it occurs at the critical point $x = 2$. Also, the absolute maximum value of f is 17 which occurs at the end point 4.

Example 2.9.2 Find the extreme values of $f(x)=\dfrac{1}{\sqrt{4-x^2}}$

Solution For

$$f(x)=\frac{1}{\sqrt{4-x^2}}=\left(4-x^2\right)^{-1/2}$$

$$f'(x)=-\frac{1}{2}\left(4-x^2\right)^{-3/2}(-2x)=\frac{x}{\left(4-x^2\right)^{3/2}}$$

The only critical point in the domain $(-2,2)$ is $x = 0$. The value $f(0)=\dfrac{1}{\sqrt{4-0^2}}=\dfrac{1}{2}$ is probable extreme value.

To determine weather $\dfrac{1}{2}$ is an extreme value of f, we examine the formula $f(x)=\dfrac{1}{\sqrt{4-x^2}}$

As x moves away from 0 on either side, the denominator gets smaller, the values of f increase and the graph rises, we have a min value at $x = 0$ which is absolute.

Exercise

1. Find the critical values of the function
   (i) $f(x)=x^3+6x^2-15x$ (ii) $g(y)=\frac{y-1}{y^2-y+1}$ (iii) $h(t)=|3t-4|$
2. Find the absolute minimum and absolute maximum of the function f on the given interval
   (i) $f(x)=12+4x-x^2,\ [0,5]$
   (ii) (ii) $f(t)=t\sqrt{4-t^2},\ [-1,2]$
   (iii) $f(x)=2cos+\sin 2x,\ \left[0,\frac{\pi}{2}\right]$
3. Find the derivative at each critical point and determine the local extreme values of the following (a) $y=x^{2/3}(x+2)$ (b) $y=x\sqrt{4-x^2}$ (c) $y=\begin{cases}4-x^2, & -2\le x<0\\ 2x-1, & 0\le x\le 2\end{cases}$

2.9.6 Derivatives and the shapes of graphs

Recall that a function f is

(i) increasing on I if for all points x_1 and x_2 in I with $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$
(ii) decreasing on I if for all points x_1 and x_2 in I with $x_2 < x_1 \Rightarrow f(x_2) < f(x_1)$

2.9.6.1 Test for increasing and decreasing function

Suppose that f is continuous on an interval $[a,b]$ and differentiable on (a, b), then

(i) f is increasing on $[a, b]$ if $f' > 0$ at each point of (a, b)
(ii) f is decreasing on $[a, b]$ if $f' < 0$ at each point of (a, b)

Example 2.9.6.1

Find the critical point of $f(x) = x^3 - 12x - 5$ and identify the intervals on which f in increasing and decreasing

Solution: Given $f(x) = x^3 - 12x - 5$

$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x-2)(x+2)$

The critical points occurs when $f'(x) = 0$ that is when $x = -2, 2$. These points partitioned the x – axis as $x < -2, -2 < x < 2, x > 2$

Interval	3	$x-2$	x+2	$f'(x)$	f
$x < -2$	+	−	−	+	Increasing on $(-\infty, -2)$
$-2 < x < 2$	+	−	+	−	Degreasing on $(-2, 2)$
$x > 2$	+	+	+	+	Increasing on $(2, \infty)$

2.9.7 The first derivative test of local extreme values

Suppose that c is a critical number of a continuous function, then

(a) f has a local minimum if f' changes from negative to positive at c.
(b) f has a local maximum if f' changes from positive to negative at c.
(c) f has no local extreme value if f' has the same sign on both sides of c.

Example 2.9.7

If $f(x) = x^{\frac{1}{3}}(x-4)$ identify the interval on which f is increasing and decreasing. Hence, find the local extreme values of f.

Solution $f'(x) = \frac{d}{dx}\left(x^{3/3} - 4x^{1/3}\right) = \frac{4}{3}x^{-\frac{2}{3}}(x-1) = \frac{4(x-1)}{3x^{\frac{2}{3}}}$

is zero at $x = 1$ and undefined at $x = 0$. The critical points partitioned the x −axis as follows:

$x < 0, 0 < x < 1, x > 1$.

Interval	$4(x-1)$	$3x^{\frac{2}{3}}$	f'	f
$x < 0$	−	+	−	Decreasing on $(-\infty, 0)$
$0 < x < 1$	−	+	−	Decreasing on $(0, 1)$
$x > 1$	+	+	+	Increasing on $(1, \infty)$

Thus, using the first derivative test for local extrema, we have:

f does not have an extreme value at $x = 0$ (f' does not change sign) and that f has a local minimum at $x = 1$ (f' changes from negative to positive). So, the value of the local minimum is $f(1) = 1 - 4 = -3.$

3.4 Concavity of a graph of *f* (*x*)

Definition: The graph of a differentiable function $y = f(x)$ is said to be

(a) Concave up on an open interval *I* if f' is increasing on *I*.

(b) Concave down on an open interval *I* if f' is decreasing on *I*.

The graph of $f(x) = x^3$ is concave down on $(-\infty, 0)$ and concave up on $(\infty, 0)$ as shown in Figure 5.

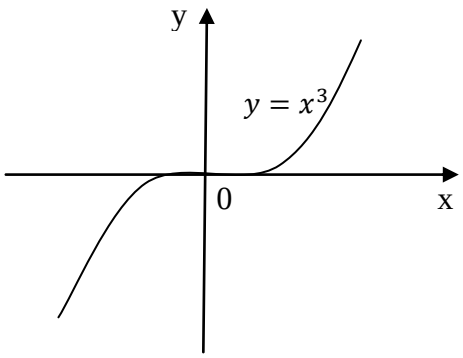


Figure 5. Concavity

3.5 Second derivative test for concavity

The graph of twice differentiable function $y = f(x)$ is

(a) Concave up on an interval *I* if $y'' > 0$ for all x in *I*.

(b) Concave down on an interval *I* if $y'' < 0$ for all x in *I*.

Example 3.5.1 Determine the concavity of $y = x^2$

Solution $y' = 2x,\ y'' = 2 > 0,$ hence $y = x^2$ is concave up.

3.6 Point of inflection

A point *P* on a curve $y = f(x)$ where f is continuous and the concavity changes is called the point of inflection.

Note that at this point y' is always zero. In view of the Concavity Test, there is a point of inflection at any point where the second derivative changes sign.

Example 3.6.1 If $y = x^4$, determine whether an inflection point exist or not.

Solution

Given $y = x^4,\ y' = 4x^3, y'' = 12x^2$ at $x = 0, y'' = 0$, then there is no point of inflection.

Example 3.6.2 Given $y = x^{1/3}$, find the point of inflection.

Solution $y = x^{1/3}, y' = \frac{1}{3}x^{-2/3}, y'' = -\frac{2}{9}x^{5/3} = \frac{-2}{9\cdot\sqrt[3]{x^5}}$ at $x = 0,\ y''$ is undefined.

Therefore $x = 0$ is the point of inflection.

3.7 Second derivative test for local extrema

Suppose $f''(x)$ is continuous near c.

(i) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$

(ii) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.

Example 3.7.1

Use the second derivative test to find the extreme value of $f(x) = x^3 - 12x - 5$

Solution

Given $f(x) = x^3 - 12x - 5$, we have

$$f'(x) = 3x^2 - 12,\ f''(x) = 6x$$

Let $f(x) = 0,$ then $3(x^2 - 4) = 0 \Rightarrow x^2 - 4 = 0 \Rightarrow x = \pm 2$

$$f''(2) = 6(2) = 12 > 0,\ f''(-2) = 6(-2) = -12 < 0.$$

Then there is a local maximum at $x = -2$ and a local minimum at $x = 2$.

Note that

If $f'' = 0$ or f'' does not exist, then the test fails and so, we go back to the first derivative test.

2.8. Curve sketching

The following are guide to sketching a curve $y = f(x)$ by hand.

(i) Determine the domain of a function $f(x)$ as it is often useful

(ii) Find the $y-$intercept ($f(0)$) and the $x-$intercepts by setting $y = 0$ and solve for x. One can omit this step if the function is difficult to solve

(iii) Check for symmetry. That is f is even or odd function.

(iv) Find asymptotes for f. There are horizontal and vertical asymptotes. A horizontal asymptote is the line $y = L$ if $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$. While a vertical asymptote is the line $x = a$ if at least one of the following limits is rue

$$\lim_{x\to a^+} f(x) = \infty, \quad \lim_{x\to a^-} f(x) = \infty$$

$$\lim_{x\to a^+} f(x) = -\infty, \quad \lim_{x\to a^-} f(x) = -\infty$$

Note that one can locate the vertical asymptote of a rational function by setting the denominator to zero after cancelling common factors.

(v) Identify where the extrema of $f(x)$ occur.

(vi) Find the interval on which f is increasing and decreasing.

(vii) Find where the graph is concave up and concave down.

(viii) Sketch the possible graph for *f*.

Example 3.8.1

Sketch the curve of the function $f(x) = \frac{2x^2}{x^2-1}$

Solution

Given $f(x) = \frac{2x^2}{x^2-1}$, we have $f'^{(x)} = \frac{4x(x^2-1)-4x^3}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2}$

and $x = 0, x = \pm 1$ are the critical points. These points portioned the $x-$axis in to three intervals $(-\infty, 0), (0, 3)$ and $(3, \infty)$.

(i) The domain of $f(x)$ is $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$

(ii) For y-intercept $f(0) = 0$ and when $y = 0, x = 0$ is the x-intercept.

(iii) For symmetry, $f(-x) = \frac{2(-x)^2}{(-x)^2-1} = f(x)$ so that f is symmetrical about the y-axis.

(iv) $\lim_{x\to\infty} \frac{2x^2}{x^2-1} = \lim_{x\to\infty} \frac{2}{1-\frac{1}{x^2}} = 2$. Thus, the line $y = 2$ is the horizontal asymptote. Since the denominator is zero at $x = \pm 1$, then $\lim_{x\to 1^+} \frac{2x^2}{x^2-1} = \infty$, $\lim_{x\to 1^-} \frac{2x^2}{x^2-1} = -\infty$,

$$\lim_{x\to -1^+} \frac{2x^2}{x^2-1} = -\infty, \lim_{x\to -1^-} \frac{2x^2}{x^2-1} = \infty.$$

Therefore the line $x = 1$ and $x = -1$ are vertical asymptotes

(v) When $x < 0$ $(x \neq -1)$ then $f' > 0$ and $f' < 0$ for $x > 0$ $(x \neq 1)$ so that f is increasing on $(-\infty, -1), (-1, 0)$ and decreasing on $(0, 1), (1, \infty)$

(vi) From the first derivative test, $f(0) = 0$ is a local maximum since f' changes sign from positive to negative at the critical point 0.

(vii) $f'' = \frac{-4(x^2-1)^2+16x^2(x^2-1)}{(x^2-1)^4} = \frac{4(x^2-1)(1+3x^2)}{(x^2-1)^4} = \frac{12x^2+4}{(x^2-1)^3}$ and $f'' > 0$ if and only if

$x^2 - 1 > 0 \Leftrightarrow |x| > 1$. Similarly, $f'' < 0 \Leftrightarrow |x| < 1$. Thus f is concave up on the intervals $(-\infty, -1)$ and $(1, \infty)$ and concave down on $(-1, 1)$. There is no point of inflection since 1 and -1 are not in the domain of f.

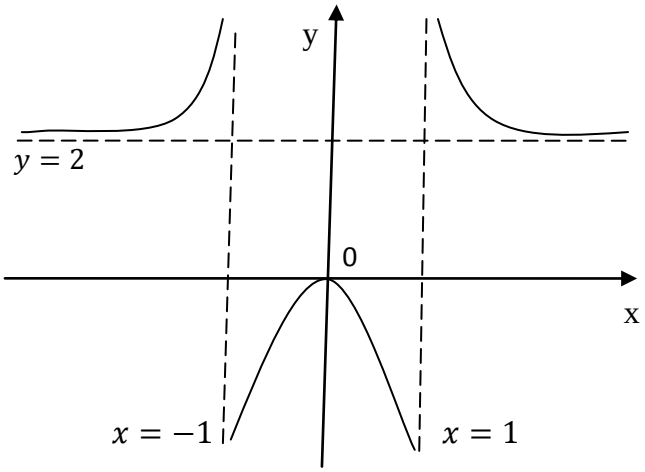


Figure 6. The curve of $f(x) = \frac{2x^2}{x^2-1}$

SECTION C

Integration

Anti-derivative of a function $f(x)$ is a function $F(x)$ such that $F'(x) = f(x)$. This is an inverse operation to differentiation which is actually called integration.

Thus, if $\frac{dy}{dx} = x^n$, then $y = \frac{x^{n+1}}{n+1}, n \neq -1$ or $y = \int x^n dx = \frac{x^{n+1}}{n+1} + c$

Where, $\int$ is the integral sign,

$\frac{x^{n+1}}{n+1}$ is the indefinite integral,

c is an arbitrary constant.

So, in general, $F(x) = \int f(x) dx$

Example 3.1 Evaluate the following integrals

(a) $\int 2x dx$ (b) $\int 4\sqrt{x} dx$ (c) $\int \frac{1}{\sqrt{x}} dx$ (d) $\int (6x^3 + 8x + 3) dx$

Solution

(a) $\int 2xdx = 2\int xdx = 2\frac{x^2}{2} + c = x^2 + c$

(b) $\int 4\sqrt{x}\,dx = 4\int \sqrt{x}\,dx = 4\int x^{\frac{1}{2}}dx = 4\frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{8}{3}x^{\frac{3}{2}} + c$

(c) $\int \frac{1}{\sqrt{x}}dx = \int x^{-½}dx = \frac{x^{½}}{\frac{1}{2}} = 2x^{½} + c = \frac{2}{\sqrt{x}} + c$

(d) $\int \left(6x^3 + 8x + 3\right)dx = 6\int x^3dx + 8\int xdx + 3\int dx = 3\frac{x^4}{2} + 4x^2 + 3x + C$

Exercise 3.1

Evaluate the following indefinite integrals.

(1) $\int \frac{dx}{x^2}$ (2) $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)dx$ (3) $\int (x+1)^3 dx$ (4) $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 dx$ (5) $\int \frac{2(x+3)}{x^3}dx$

3.1 Rules for indefinite integrals

Let f be a function and c a real number, then

(1) $\int cf(x)dx = c\int f(x)dx$, where *A* is a constant.

(2) $\int -f(x)dx = -\int f(x)dx$

(3) $\int \left(f(x) \pm g(x)\right)dx = \int f(x)dx \pm \int g(x)dx$

When a function $f(x)$ is integrated over a limit say *a*, *b*, the integral is given by:

$$\int_a^b f(x)dx = F(b) - F(a),$$

where, *a*, is the lower limit and *b* is the upper limit. This is called a definite integral. Such type of integrals have the following properties.

(i) $\int_a^a f(x)dx = 0$

(ii) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

Example 3.1 Evaluate the following (1) $\int_1^3 (x^2 - 1)dx$ (2) $\int_1^2 \left(3 - \frac{6}{x^2}\right)dx$

Solution (1) $\int_1^3 (x^2 - 1)dx = \int_1^3 x^2dx + \int_1^3 dx = \frac{x^3}{3} + x\Big|_1^3 = \left(\frac{27}{3} - \frac{1}{3}\right) + (3-1) = \frac{32}{3}$

(2) $\int_1^2 \left(3 - \frac{6}{x^2}\right)dx = 3\int dx - 6\int x^{-2}dx = 3x + \frac{6}{x}\Big|_1^2 = 6 - 3 + 3 - 6 = 0$

Exercise 3.1

Evaluate the following definite integrals

(a) $\int_0^1 dx$ (b) $\int_a^{2a}(x+1)(x-2)dx$ (c) $\int_4^9 \frac{dx}{\sqrt{x}}$ (d) $\int_0^3(6x^2-4x+1)dx$ (e) $\int_3^6\left(x+\frac{1}{x^2}\right)dx$

3.2 Integration by substitution

The substitution method is used to evaluate the integrals that cannot be integrated directly and a function and its derivative exist in the integrand. That is, this method works whenever we have an integral of the form

$$\int f(g(x))g'(x)dx.$$

Example3.2 Evaluate (1) $\int\sqrt{2x+1}dx$ (2) $\int x(3x^2+2)^4dx$

Solution

(1) Let $u=2x+1,\ du=2dx,$ then

$$\int\sqrt{2x+1}dx=\frac{1}{2}\int u^{1/2}du=\frac{1}{2}\frac{u^{3/2}}{3/2}+c=\frac{1}{3}(2x+1)^{3/2}+c$$

(2) Let $u=3x^2+2,\ du=6xdx,$ then

$$\int x(3x^2+2)^4dx=\int \cancel{x}\cdot u^4\frac{du}{6\cancel{x}}=\frac{1}{6}\int u^4du$$

$$=\frac{1}{6}\left(\frac{u^5}{5}\right)=\frac{1}{30}(3x^2+2)^4+C$$

For definite integral if g' is continuous on a closed interval $[a,b]$ and f is continuous on the range of $u=g(x)$, then

$$\int_a^b f(g(x))\cdot g'(x)dx=\int_{g(a)}^{g(b)}f(u)du.$$

We use this by putting $u=g(x),\ du=g'(x)dx$ and integrate from $g(a)$ to $g(b)$.

Example 3.2.1 Use the substitution method to evaluate $\int_{-1}^1 3x^2\sqrt{x^3+1}dx$

Solution Let $u=x^3+1,\ du=3x^2dx$ and when $x=-1, u=0$ while $u=2$ for $x=1$, then

$$\int_{-1}^1 3x^2\sqrt{x^3+1}dx=\int_0^2\sqrt{u}du=\left[\frac{2}{3}u^{3/2}\right]_0^2=\frac{2}{3}\left[2^{3/2}-0\right]=\frac{4\sqrt{2}}{3}$$

Exercise 3.2

(1) Evaluate the following integrals by substitution.

(a) $\int \frac{3rdr}{\sqrt{1-r^2}}$ (b) $\int t^2\left(1+2t^3\right)^{-2/3}dt$ (c) $\int \frac{z+1}{3\sqrt{z^2+2z+2}}dz$

(d) $\int \frac{(x+1)}{\sqrt{x^2+2x-4}}dx$ (e) $\int \frac{x}{\left(x^2+4\right)^3}dx$ (f) $\int_1^4 \frac{dy}{2\sqrt{y}\left(1+\sqrt{y}\right)^2}$

(2) Evaluate the following integrals.

(a) $\int_{-1}^{1} \frac{5r}{\left(4+r^2\right)^2}dr$ (b) $\int_0^1 \sqrt{t^5+2t}\left(5t^4+2\right)dt$ (c) $\int_0^1 \frac{x^2-1}{\sqrt{x^3-3x}}dx$

(d) $\int_0^1 \left(4y-y^2+4y^3+1\right)^{-2/3}\left(12y^2-2y+4\right)dy$

3.3 Integration of symmetric functions

Suppose f is continuous function on an interval $[-a, a]$,

(i) If f is even, then $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$

(ii) If f is odd, then $\int_{-a}^{a} f(x)dx = 0$

Example 3.3 Evaluate (i) $\int_{-2}^{2}(x^6+1)dx$ (ii) $\int_{-1}^{1} \frac{\tan x}{x^4+x^2+1}dx$

Solution

(i) $f(x) = x^6 + 1$ is an even function since it satisfies $f(-x) = f(x)$ and so,

$$\int_{-2}^{2}(x^6+1)dx = 2\int_0^2 (x^6+1)dx = 2\left(\frac{x^7}{7}+x\right)\Big|_0^2 = \frac{256}{7}+4 = \frac{284}{7}$$

(ii) $f(x) = \frac{\tan x}{x^4+x^2+1}$ is an odd function as $f(-x) = \frac{\tan(-x)}{(-x)^4+(-x)^2+1} = \frac{-\tan x}{x^4+x^2+1} = -f(x)$ and so

$$\int_{-1}^{1} \frac{\tan x}{x^4+x^2+1}dx = 0.$$

Exercise 3.3

3.4 Integration of trigonometric functions and substitution

Recall that $\frac{d}{du}(\sin u) = \cos u\frac{du}{dx}$ and $\frac{d}{du}(\cos u) = -\sin u\frac{du}{dx}$ imply

$$\int \cos u\, du = \sin u + c \text{ and } \int \sin u\, du = -\cos u + c$$

Similarly, we have

$$\int \sec^2 u\, du = \tan u + c \text{ and } \int \csc^2 u\, du = -\cot u + c$$

Example 3.4.1

Evaluate (1) $\int \sin 2x dx$ (2) $\int \frac{\cos 2x}{\sin^3 2x}dx$ (3) $\int \sec^2 5x dx$ (4) $\int \cos(7\theta+5)d\theta$

Solution

(1) Let $u = 2x \Rightarrow du = 2dx$ then $\int \sin 2x dx = \int \sin u \frac{du}{2} = -\frac{1}{2}\cos 2x + c$

(2) Let $u = \sin 2x \Rightarrow du = 2\cos 2x dx$ and $dx = \frac{du}{2\cos 2x}$ then

$$\int \frac{\cos 2x}{\sin^3 2x} dx = \int \frac{\cos 2x}{u} \cdot \frac{du}{2\cos 2x} = \frac{1}{2}\int \frac{du}{u^3} = -\frac{1}{4\sin^2 2x} + C$$

(3) $\int \sec^2 5x dx = \frac{1}{5}\tan 5x + c$

(4) $\int \cos(7\theta + 5) d\theta = \frac{1}{7}\sin(7\theta + 5) + C$

3.4.1 Odd powers of trigonometric functions

In this case, we use the identity $\cos^2 x + \sin^2 x = 1$ to transform integrands to the form of a function and its derivative and use the substitution method to integrate.

Example

Evaluate (1) $\int \cos^3 x dx$ (2) $\int \sin^5 2x dx$

(1) $\int \cos^3 x dx = \int \cos x \cdot \cos^2 x dx = \int \cos x (1 - \sin^2 x) dx = \sin x - \int \cos x \sin^2 x dx$

Let $u = \sin x$, $du = \cos x dx$ then

$$\int \cos^3 x dx = \sin x - \int \cos x \cdot u^2 \frac{du}{\cos x} = \sin x - \frac{\sin^3 x}{3} + C$$

(2) $\int \sin^5 2x dx = \int \sin 2x \cdot \sin^4 2x dx$

$$= \int \sin 2x (sin^2 2x)^2 \, dx = \int \sin 2x (1 - cos^2 2x)^2 \, dx$$

$$= \int \sin 2x - 2 \sin 2x \, cos^2 2x + \sin 2x \, cos^4 2x \, dx$$

$$= -\frac{\cos 2x}{3} + \frac{cos^3 2x}{3} - \frac{cos^5 2x}{10} + c$$

3.4.2 Even powers of trigonometric functions

In this case the following identities may be used to evaluate such integrals.

(i) $\cos^2 x + \sin^2 x = 1$ (ii) $\cos^2 x - \sin^2 x = \cos 2x$

(iii) $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ (iv) $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

Example

Evaluate (1) $\int \cos^2 x dx$ (2) $\int \cos^4\left(\frac{x}{2}\right) dx$

Solution

(1) $\int \cos^2 x dx = \int \frac{1}{2}(1+\cos 2x) dx = \frac{1}{2}\int (1+\cos 2x)\, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$

(2) $\int \cos^4\left(\frac{x}{2}\right) dx = \int \left(\cos^2\left(\frac{x}{2}\right)\right)^2 dx = \int \frac{1}{2}(1+\cos x)^2\, dx = \frac{1}{2}\int (1+2\cos x + \cos^2 x) dx$

$$= \frac{1}{2}(x + 2\sin x) + \frac{1}{2}\int \frac{1}{2}(1+\cos 2x) dx$$

$$= \frac{x}{2} + \sin x + \frac{1}{2}\left(\frac{x}{2} + \frac{\sin 2x}{2}\right) + c = \frac{1}{2}\left(\frac{x}{2} + 2\sin x + \frac{x}{2} + \frac{\sin 2x}{2}\right) + c$$

Exercise 3.4

Evaluate the following

(1) $\int \left(1-\sin^2 3t\right)\cos 3t dt$ (2) $\int \frac{\sin x}{\cos^2 x} dx$ (3) $\int \frac{\sin 2t dt}{\sqrt{2-\cos 2t}}$ (4) $\int \cot^3 x dx$

(5) $\int \sec^4 x dx$ (6) $\int \csc^3 x \cot x dx$ (7) $\int \sec^2 x \tan^2 x dx$

(8) Find $\int \tan x \sec^4 x dx$, (i) as a function of $\sec x$ (ii) as a function of $\tan x$, and show that they are the same.

3.5 Integration of exponential functions

3.5.1Natural exponentials

Recall that $\frac{d}{dx}e^u = e^u \frac{du}{dx}$ and so $\int e^u du = e^u + c$

Example 3.5.1

Find (1) $\int e^{x/2} dx$ (2) $\int x e^{x^2} dx$ (3) $\int_0^{\pi/2} e^{\sin x} \cos x dx$

Solution

(1) $\int e^{x/2} dx = 2e^{x/2} + c$

(2) Let $u = x^2$, $du = 2x dx$ then $\int x e^{x^2} dx = \frac{1}{2}e^u + c = \frac{1}{2}e^{x^2} + c$

(3) $\int_0^{\pi/2} e^{\sin x} \cos x dx = e^{\sin x}\Big|_0^{\pi/2} = e^1 - e^0 = e - 1$

3.5.2 General exponential functions

Recall that $\frac{d}{dx}a^u = a^u \ln a \frac{du}{dx}$

Dividing through by $\ln a$, we have $a^u \frac{du}{dx} = \frac{1}{\ln a}\frac{d}{dx}(a^u)$

Integrating this with respect to x, we get

$$\int a^u \frac{du}{dx}dx = \int \frac{1}{\ln a}\frac{d}{dx}(a^u)dx = \frac{1}{\ln a}\int \frac{d}{dx}(a^u)dx = \frac{1}{\ln a}a^u + c$$

Therefore

$$\int a^u du = \frac{a^u}{\ln a} + c$$

Example

Evaluate (1) $\int 3^x dx$ (2) $\int 2^{\sin x}\cos x dx$

Solution

(1) $\int 3^x dx = \frac{3^x}{\ln 3} + c$

(2) Let $u = \sin x, \quad du = \cos x dx$ then

$$\int 2^{\sin x}\cos x dx = \int 2^u du = \frac{2^u}{\ln 2} + c = \frac{2^{\sin x}}{\ln 2} + c$$

Exercise 3.5.2

Evaluate the following integrals

(1) $\int \left(e^{3x} + 5e^{-x}\right)dx$ (2) $\int \frac{e^{-\sqrt{r}}}{\sqrt{r}}dr$ (3) $\int_0^{\pi/4}\left(1+e^{\tan\theta}\right)\sec^2\theta d\theta$

(4) $\int \frac{e^{\frac{1}{x}}}{x^2}dx$ (5) $\int \sin x e^{\cos x}dx$ (6) $\int_1^{\sqrt{2}} 2^{x^2}\cdot x dx$

(7) $\int 8e^{(x+1)}dx$ (8) $\int_1^2 \frac{2^{\ln x}}{x}dx$ (9) $\int x^{\sqrt{2}-1}dx$

(10) $\int_1^e x^{(\ln 2)-1}dx$

3.6 Method of changing variable

This method is used when there is a function without its derivative in a given integrand.

Example (1) $\int 3x\sqrt{4x-1}dx$ (2) $\int_{x=\frac{1}{2}}^{x=3} x\sqrt{2x+3}dx$

Solution

(1) Let $u = \sqrt{4x-1} \Rightarrow u^2 = 4x-1 \Rightarrow x = \frac{1}{4}\left(u^2+1\right)$ and $dx = \frac{1}{4}(2u)du = \frac{u}{2}du$

$$\therefore \int 3x\sqrt{4x-1}dx = \int 3\left(\frac{1}{4}\left(u^2+1\right)\right)\cdot u\cdot\frac{u}{2}du = \frac{3}{8}\int\left(u^4+u^2\right)du = \frac{3}{8}\left(\frac{u^5}{5}+\frac{u^3}{3}\right)+c$$

$$= \frac{u^3}{40}(3u^2+5)+c = \frac{\left(\sqrt{4x-1}\right)^3}{40}(12x+2)+c = \frac{\left(\sqrt{4x-1}\right)^3}{20}(6x+1)+c$$

(2) Let $2x+3=u \Rightarrow x=\frac{1}{2}(u-3)$, and $dx=\frac{1}{2}du$

$$\int_{½}^{3} x\sqrt{2x+3}dx = \int_4^9\left(\frac{1}{4}u^{3/2}-\frac{3}{4}u^{1/2}\right)du = \left[\frac{u^{5/2}}{10}-\frac{u^{3/2}}{2}\right]_4^9 = \frac{116}{10}$$

Exercise 3.5

Evaluate (1) $\int x\sqrt{5x+2}dx$ (2) $\int x(2x-1)^6\,dx$ (3) $\int_{½}^{3} x\sqrt{2x+3}dx$

3.6 Logarithmic functions

3.6.1 Natural Logarithm

Recall that $\frac{d}{dx}(\ln u)=\frac{1}{u}du$, then $\int\frac{1}{u}du=\ln|u|+c=\ln|ku|$, where $c=\ln k$

Example

Evaluate the following: (1) $\int\frac{1}{2x}dx$ (2) $\int\frac{1}{2x-1}dx$ (3) $\int_{-\pi/2}^{\pi/2}\frac{4\cos\theta}{3+2\sin\theta}d\theta$

Solution:

(1) $\int\frac{1}{2x}dx=\frac{1}{2}\int\frac{1}{x}dx=\frac{1}{2}\ln x+c=\ln k\sqrt{x}$

(2) $\int\frac{1}{2x-1}dx=\frac{1}{2}\ln(2x-1)+c=\ln k\sqrt{2x-1}$

(3) Let $u=3+2\sin\theta\Rightarrow du=2\cos\theta d\theta$ and $u(-\pi/2)=1,\ u(\pi/2)=5$

$$\int_{-\pi/2}^{\pi/2}\frac{4\cos\theta}{3+2\sin\theta}d\theta=\int_1^5\frac{2}{u}du=\left[2\ln u\right]_1^5=2\ln 5$$

3.6.2 Logarithm to base *a*

To evaluate integrals involving logarithm to base *a*, we need to convert them to natural logarithms as follows

$$\log_a f(x)=\frac{\ln f(x)}{\ln a}.$$

Example Find (1) $\int\frac{\log_2 x}{x}dx$ (2) $\int_1^2\frac{\log_2(x+2)}{x+2}dx$

Solution

(1) $\log_2 x = \dfrac{\ln x}{\ln 2}$

Let $u = \ln x,\ du = \dfrac{1}{x}dx$

$$\int \frac{\log_2 x}{x} = \frac{1}{\ln 2}\int \frac{\ln x}{x}dx = \frac{1}{\ln 2}\int u du = \frac{1}{\ln 2}\frac{u^2}{2} + C = \frac{(\ln x)^2}{2\ln 2} + C$$

(2) $\log_2(x+2) = \dfrac{\ln(x+2)}{\ln 2}$

Let $u = \ln(x+2) \Rightarrow du = \dfrac{1}{x+2}dx$

$$\int_1^2 \frac{\log_2(x+2)}{x+2}dx = \int_1^2 \frac{1}{x+2}\cdot\frac{\ln(x+2)}{\ln 2}dx = \int_1^2 \frac{u}{\ln 2}du = \frac{1}{\ln 2}\left(\ln|u|_1^2\right) = \frac{1}{\ln 2}\cdot \ln 2 = 1$$

Exercise 3.6

Evaluate the following integrals

(1) $\int_{-3}^{-2}\dfrac{dx}{x}$ (2) $\int \dfrac{2y}{y^2-25}dy$ (3) $\int_0^{\pi}\dfrac{\sin t}{2-\cos t}dt$ (4) $\int_1^2 \dfrac{2\ln x}{x}dx$ (5)

$\int_2^4 \dfrac{dx}{x(\ln x)^2}$ (6) $\int \dfrac{3\sec^2 t}{6+3\tan t}dt$ (7) $\int_0^{\pi/2}\tan\frac{x}{2}dx$ (8) $\int_{\pi/2}^{\pi} 2\cot\frac{\theta}{3}d\theta$

(9) $\int \dfrac{dx}{2\sqrt{x}+2x}$ (10) $\int \dfrac{\log_{10} x}{x}dx$ (11) $\int \dfrac{dx}{x\log_{10} x}$

3.7 Integration by method of partial fractions

Recall that a rational function $f(x) = \frac{P(x)}{Q(x)}$ which is proper ($\deg(P) < \deg(Q)$) can be resolved into the sum of simpler fractions. Moreover, if $f(x)$ is improper ($\deg(P) \geq \deg(Q)$) we must take the preliminary step of long division such that

$$f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

where S and R are polynomials

3.7.1 Denominator with linear factors

In this case, $f(x)$ can be expressed in the form $f(x) = \frac{P(x)}{Q(x)} = \frac{A_1}{a_1x+b_1} + \frac{A_2}{a_2x+b_2} + \cdots + \frac{A_k}{a_kx+b_k}$ as in the following examples.

Example (1) $\int \frac{dx}{x(x-2)}$ (2) $\int \frac{x-2}{(x-5)(x+1)}dx$

Solution

(1) $\frac{1}{x(x-2)} = \frac{A}{x} + \frac{B}{x-2} = \frac{A(x-2)+Bx}{x(x-2)}$ which imply

$$1 = A(x-2) + Bx \Rightarrow A = \frac{1}{2}, B = \frac{1}{2}$$

Therefore

$$\int \frac{1}{x(x-2)}\,dx = \int \left(\frac{\frac{1}{2}}{x} + \frac{\frac{1}{2}}{x-2} \right) dx = \ln k\sqrt{x(x-2)}$$

3.7.2 Denominator with repeated factor

If the first linear factor in the factorization of $Q(x)$ is repeated r times then

$$f(x) = \frac{P(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}$$

Example Find (a) $\int \frac{5x+2}{(x-2)^2(x+1)}\,dx$ (b) $\int \frac{2x-1}{(x+1)^2}\,dx$

Solution

(a) $$\frac{5x+2}{(x-2)^2(x+1)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+1}$$

$$\Rightarrow 5x+2 = A(x+1)(x-2) + B(x+1) + C(x-2)^2$$

put $x = 2$, $B = 4$, put $x = -1$, $C = -\frac{1}{3}$, putting $x = 0$, $A = \frac{1}{3}$

$$\frac{5x+2}{(x-2)^2(x+1)} = \frac{1}{3(x-2)} + \frac{4}{(x-2)^2} - \frac{1}{3(x+1)}$$

$$\int \frac{5x+2}{(x-2)^2(x+1)}\,dx = \frac{1}{3}\int \frac{1}{x-2}\,dx + 4\int \frac{1}{(x-2)^2}\,dx - \frac{1}{3}\int \frac{1}{x+1}$$

$$= \tfrac{1}{3}\ln(x-2) + \tfrac{4}{x-2} - \tfrac{1}{3}\ln(x+1) = \ln(x-2)^{\frac{1}{3}} + \tfrac{4}{x-2} - \ln(x+1)^{\frac{1}{3}}$$

$$= \ln k\sqrt[3]{\frac{x-2}{x+1}} - \frac{4}{x-2}$$

3.7.3 Denominator with non repeated quadratic factors

Here, if $Q(x)$ has quadratic irreducible factor $ax^2 + bx + c$, then the expression for $f(x)$ will have a term of the form $\frac{Ax+B}{ax^2+bx+c}$, where A and B are constants to be determined.

Example Evaluate (i) $\int \frac{7x+2}{3x^3+x^2}\,dx$ (ii) $\int \frac{2x^2+2x+3}{(x+2)(x^2+3)}\,dx$

Solution

(i) $\int \frac{7x+2}{3x^3+x^2}\,dx = \int \frac{7x+2}{x^2(3x+1)}\,dx$

But $\frac{7x+2}{3x^3+x^2} = \frac{Ax+B}{x^2} + \frac{C}{3x+1} \Rightarrow 7x+2 = (Ax+B)(3x+1) + x^2C$

put $x=0$, $B=2$, put $x=-\frac{1}{3}$, $C=-3$, put $x=1$, $A=1$

$\int \frac{7x+2}{3x^3+x^2}dx = \int \left(\frac{x+2}{x^2} - \frac{3}{3x+1}\right)$

$$= \ln x - \frac{2}{x} - \ln(3x+1) + C = \ln k\left(\frac{x}{3x+1}\right) - \frac{2}{x}$$

For the improper fractions, we apply the usual long division method and then use one of the previous three cases on the resulting expression as in the example below.

Example: Evaluate $\int \frac{x^3-18x-21}{(x+2)(x-5)}dx$

Solution

$$\int \frac{x^3-18x-21}{(x+2)(x-5)}dx = \int \frac{x^3-18x-21}{x^2-3x-10}dx$$

$$\begin{array}{r} x+3 \\ x^2-3x-10 \overline{)\, x^3-18x-21} \\ \underline{x^3-3x^2-10x} \\ 3x^2-8x-21 \\ \underline{3x^2-9x-30} \\ x+9 \end{array}$$

Thus

$$\frac{x^3-18x-21}{(x+2)(x-5)} = x+3+\frac{x+9}{(x+2)(x-5)}$$

and so,

$$\int \frac{x^3-18x-21}{(x+2)(x-5)}dx = \int (x+3)dx + \int \frac{x+9}{(x+2)(x-5)}dx$$

But

$$\int \frac{x+9}{(x+2)(x-5)}dx = \int \left(\frac{A}{x+2} + \frac{B}{x-5}\right)dx$$

$$= -\ln(x+2) + 2\ln(x-5) + C$$

$$\int \frac{x^3-18x-21}{(x+2)(x-5)}dx = \frac{x^2}{2}+3x-\ln(x+2)+\ln(x-5)^2+C$$

$$= \frac{x^2}{2}+3x+\ln\left[\frac{(x-5)^2}{x+2}\right]+C$$

Exercise 3.7

Find the following integrals

(1) $\int \frac{1}{(x+3)(5x-2)}dx$ (2) $\int \frac{xdx}{16-x^2}$ (3) $\int \frac{x-2}{x\ -4x-5}dx$

(4) $\int \frac{2x^2+2x+3}{(x+2)(x^2+3)}dx$ (5) $\int \frac{22-16x}{(3+x)(2-x)(4-x)}dx$ (6) $\int \frac{5x+2}{(x-2)^2(x+1)}dx$

(7) $\int \frac{x^2-8x+5}{(2x+1)(x^2+9)}dx$ (8) $\int \frac{6-9x}{27x^3+8}dx$

3.8 Integration by parts

Let u and v be differentiable functions of x, then by the product rule $\frac{d}{dx}(uv)=u\frac{dv}{dx}+v\frac{du}{dx}$

implying that

$$\int\left(u\frac{dv}{dx}\right)dx = \int\left(\frac{d}{dx}(uv)\right)dx - \int\left(v\frac{du}{dx}\right)dx$$

$$\Rightarrow \int udv = uv - \int vdu$$

This gives the relationship for the integral of the product of two functions *u* and *v*.

Example

Evaluate (1) $\int x\cos xdx$ (2) $\int x^2e^x dx$

Solution

(1) Let $u = x,\ \ dv = \cos xdx \Rightarrow v = \sin x$

$$\therefore \int x\cos xdx = x\sin x - \int \sin xdx = x\sin x + \cos x + C$$

(2) Let $u = x^2,\quad dv = e^x dx \Rightarrow v = e^x$

$$\int x^2e^x dx = x^2e^x - 2\int xe^x dx$$

Let $u = x \Rightarrow du = dx,\ \ dv = e^x dx \Rightarrow v = e^x$

$$\int xe^x dx = xe^x - \int e^x dx$$

$$= xe^x - e^x + C$$

$$\therefore \int x^2 e^x dx = x^2 e^x - 2xe^x + 2e^x + C$$

Exercise 3.8

Use integration by parts to evaluate the following.

(1) $\int x \sin \frac{x}{2} dx$ (2) $\int t^2 \cos t dt$ (3) $\int_1^2 x \ln x dx$ (4) $\int x \sec^2 x dx$

(5) $\int x^3 e^x dx$ (6) $\int (x^2 - 5x) e^x dx$ (7) $\int x^5 e^x dx$ (8) $\int_0^{\pi/2} \theta^2 \sin 2\theta d\theta$

(9) $\int_0^{\pi/2} x \cos x dx$ (10) $\int_0^1 x^2 e^2 dx$

3.9 Application of integrations

To find the area bounded by the curve $y = f(x)$ above, the curve $y = g(x)$ below and on the left and right by the lines $x = a$ and $x = b$, we integrate the functions that define the region.

We first approximate the region with n −vertical rectangles by partition.

$$p = \{x_0, x_1, ..., x_n\} \text{ of } [a,b]$$

Then the area of the *kth* rectangle

$$\Delta A_k = \text{height} \times \text{width}$$

$$= [f(x_k) - g(x_k)]\Delta x_k$$

The area of the region is approximated by adding the area of the n– rectangles.

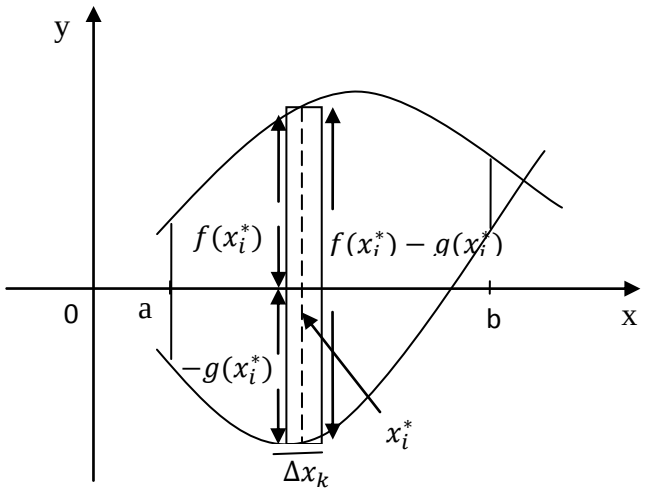


Figure 7. Typical rectangle

$$A = \sum_{k=1}^{n} \Delta A_k = \sum_{k=1}^{n} [f(x_k) - g(x_k)]\Delta x_k$$

By the Riemann sum as $\|p\| \to 0$ the sum on the right approaches the limit $\int_a^b [f(x) - g(x)]dx$

$$A = \lim_{\|p\| \to 0} \sum [f(x) - g(x)]\Delta x_k$$
$$= \int_a^b [f(x) - g(x)]dx$$

Where $\Delta A_k =$ area of kth rectangles

$f(x_k) - g(x_k) =$ Width

3.9.1 Area between curves

If f and g are continuous with $f(x) \geq g(x)$ throughout [a, b], then the area A bounded by the curves $y = f(x)$, $y = g(x)$ and the lines $x = a$ and $x = b$ is the definite integral of $(f - g)$ over the limit a to b. That is

$$A = \int_a^b [f(x) - g(x)]dx$$

Example 3.9.1 Find the area enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

Solution The region is shown in Figure

$y = 2 - x^2, y = -x \Rightarrow 2 - x^2 = -x \Rightarrow x = -1, x = 2$

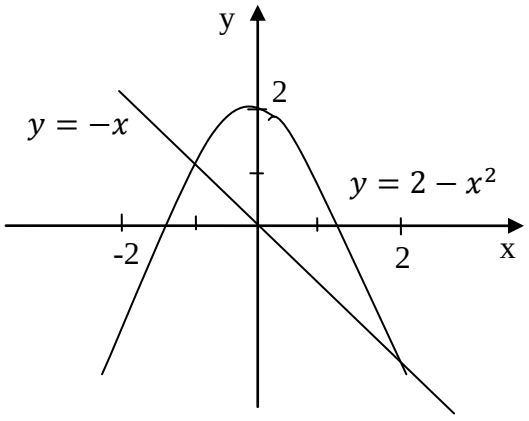


Figure 8. The given region

The region runs from $x = -1$ to $x = 2$. The limits of integration are: $a = -1, b = 2$

Now, $f(x) - g(x) = 2 - x^2 - (-x) = 2 - x^2 + x$, so that

$$A = \int_a^b [f(x) - g(x)]dx$$

$$= \int_{-1}^{2} (2 + x - x^2)dx = 2x + \frac{x^2}{2} - \frac{x^3}{3}\Big|_{-1}^{2}$$

$$= \left(6 - \frac{8}{3}\right) - \left(-2 + \frac{1}{2} + \frac{1}{3}\right) = \frac{10}{3} + \frac{7}{6} = \frac{27}{6} = \frac{9}{2}$$

Thus the area is $\frac{9}{2}$ square unit.

Example 3.9.2

Find the area bounded above by the curve $y = e^x$, bounded below by $y = x^x$ and bounded by the sides by $x = 0$ and $x = 1$.

Solution The given region is depicted in Figure 9. Then the area is given by

$$\int_0^1 (e^x - x)\, dx = e^x - \frac{x^2}{2}\Big|_0^1 = e - \frac{1}{2} - 1 = e - 1.5$$

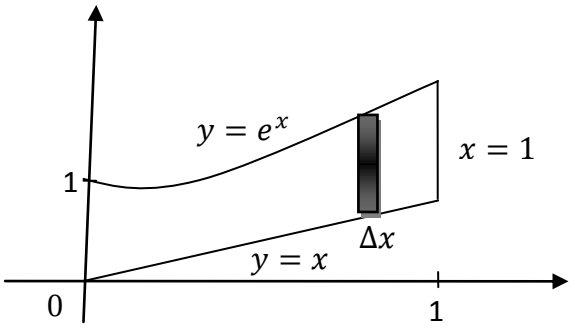


Figure 9

Exercise 3.9.1

(1) Find the area enclosed by the curves $y = 2x^2$ and $y = 12x^2 - x^3$.

(2) Sketch the curve $y = 3x - x^2$ and calculate the area between the curve and the x- axis.

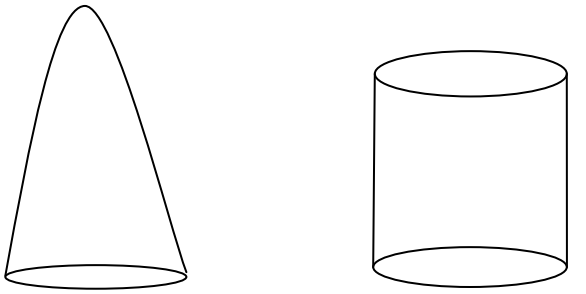
(3) Find the area under $y = 4x^3 + 8x^2$ from $x = -2$ to $x = 0$.

(4) Find the area between the curve $y = \sin x$ and the x – axis from $x = 0$ to $x = \pi$

(5) Find the area enclosed between the curves $y = 2x^2$ and $y^2 = 4x$.

3.9.2 Solids of Revolution

A solid with the central axis of symmetry is called the solid of revolution. E.g., a cone, cylinder, flower base etc. Such a solid can usually obtained by rotating a given region either about the x −axis or y-axis.



Cone Circular cylinder

Example 1 Find the volume of the solid obtained by rotating about the x-axis the region under the curve $y = \sqrt{x},\ 0 \le x \le 1$ and the x – axis.

Solution The given region is depicted in Figure 10 (a) and when we rotate about the x-axis, we obtain the solid in 10 (b). If we slice through the point x, we get a disk with radius $\sqrt{x}$.

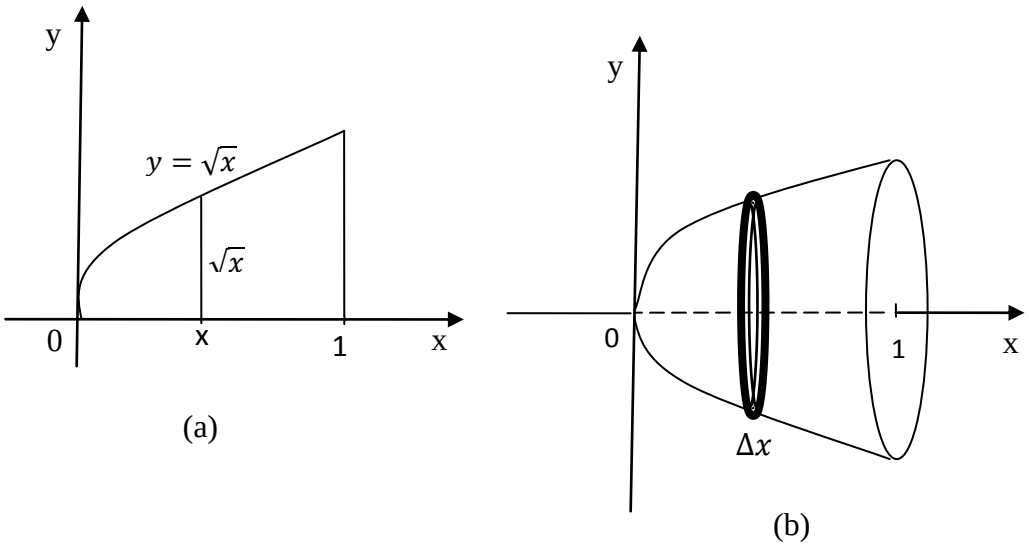


Figure 10

The area of this cross-section is $A(x) = \pi\left(\sqrt{x}\right)^2 = \pi x$. Then the volume of the approximating cylinder (a disk with thickness Δx) is $A(x)\Delta x = \pi x \Delta x$. Since the solid lies between $x = 0$ and $x = 1$, the volume is

$$V = \int_0^1 A(x)\,dx = \int_0^1 \pi x\,dx = \pi\frac{x^2}{2}\Big|_0^1 = \frac{\pi}{2} \text{ unit cube}$$

Example 2 Find the volume of the solid obtained by rotating the region bounded by $y = x^3, y = 8$, and $x = 0$ about the y-axis.

Solution

The given region is shown in Figure 11 (a) and the corresponding solid in 11 (b).

If we sliced at height y, we obtain a circular disk with radius x, where $x = \sqrt[3]{y}$. Then the area of a cross-section through y is

$$A(y) = \pi x^2 = \pi(\sqrt[3]{y})^2 = \pi y^{\frac{2}{3}}$$

Thus the volume of the approximating cylinder as in 11(b) is $A(y)\Delta y$ and so

$$V = \int_0^8 \pi y^{\frac{2}{3}}\,dy = \pi\left[\frac{3}{5}y^{\frac{5}{3}}\right]_0^8 = \frac{96\pi}{5}$$

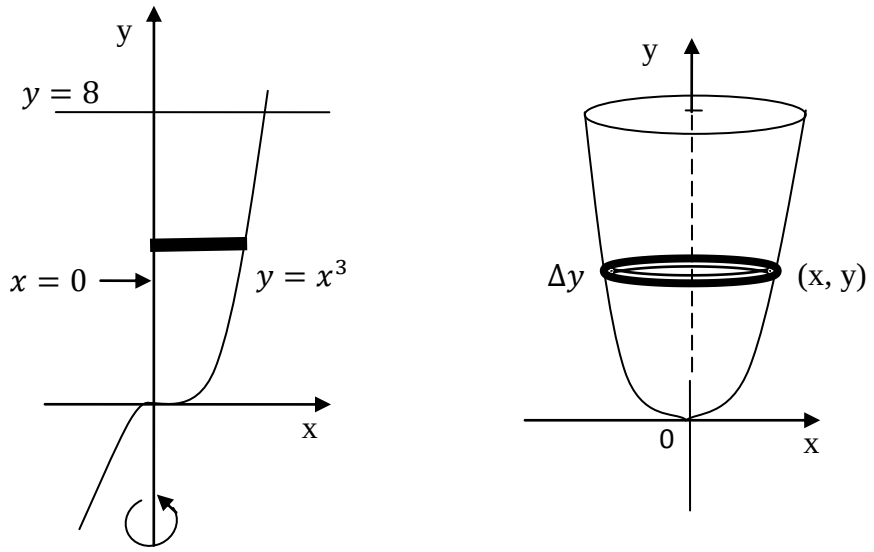


Figure 11

Exercise 1.9

1. Sketch the region enclosed by the given curves and find its area
   (a) $y = 12 - x^2, y = x^2 - 6$
   (b) $y = \cos x\,, y = 2 - \cos x,\ 0 \le x \le 2\pi$
   (c) $y = \frac{1}{x}, y = x, y = \frac{x}{4}, x > 0$
   (d) $y = e^x, y = xe^x, x = 0$
2. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line.
   (i) $y = x^3, y = x, x \ge 0$ about the $x-$axis

(ii) $y^2 = x, x = 2y$ about the $y-$axis

(iii) $y = x, y = \sqrt{x}$ about the $y-$axis

(iv) $y = x\ y = \sqrt{x}$ about $x = 2$